FORMULAS FROM PDFs 01-09

Original notation and teaching order | Read down each column; read each row from left to right

PDF 01 / slide 03
$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$

PDF 01 / slide 04
$X = \{x_1, x_2, \ldots, x_n\} \qquad J(c)$

$J(c) = \sum_{i=1}^{n}(x_i - c)^2 \qquad \bar{x} = \operatorname{argmin}_c \sum_{i=1}^{n}(x_i - c)^2$

PDF 01 / slide 05
$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Greater Error Path: $\|\mathbf{x} - c \cdot \mathbf{1}_n\|^2$

Error: $e = \mathbf{x} - P_L(\mathbf{x}) \qquad c \cdot \mathbf{1}_n$

Direction Vector: $\mathbf{1}_n = (1, 1, 1)^T$

$P_L(\mathbf{x}) = \bar{x} \cdot \mathbf{1}_n$

Optimal Scalar $\bar{x} = \operatorname{argmin}_c \|\mathbf{x} - c \cdot \mathbf{1}_n\|^2$

Geometric form: $P_L(\mathbf{x}) = \bar{x}\mathbf{1}_n = \left(\frac{\mathbf{x} \cdot \mathbf{1}_n}{\|\mathbf{1}_n\|^2}\right)\mathbf{1}_n$

Detailed Scalar: $\bar{x} = \frac{\mathbf{x} \cdot \mathbf{1}_n}{n} = \frac{\sum x_i}{n}$

PDF 01 / slide 06
$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)} \qquad \operatorname{Med}(X)$

$\tilde{X} = \begin{cases} X_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd,} \\ \frac{X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)}}{2}, & \text{if } n \text{ is even.} \end{cases}$

PDF 01 / slide 07
$\tilde{X} = \operatorname{argmin}_{m \in \mathbb{R}} \sum_{i=1}^{n} |x_i - m|$

PDF 01 / slide 08
$R = 1 \qquad d_{L2} = \sqrt{x^2 + y^2 + z^2} \le R$

$d_{L1} = |x| + |y| + |z| \le R$

$d_{L2} = \sqrt{(3-0)^2 + (2-0)^2 + (4-0)^2} = \sqrt{9+4+16} = \sqrt{29}$

$d_{L2} \approx 5.39$

$d_{L1} = |3-0| + |2-0| + |4-0| = 3+2+4$

$d_{L1} = 9$

PDF 01 / slide 09
$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{6+3+9}{3} = 6$

$\tilde{x} = 6 \qquad \bar{x} = \frac{6+3+12}{3} = 7 \qquad \tilde{x} = 6$

PDF 01 / slide 11
Median (Robust Center):	1.05
Arithmetic Mean (Expected Value):	1.53

PDF 01 / slide 12
mean > median   mean < median

PDF 01 / slide 15
$X = \{x_1, x_2, \ldots, x_n\} \qquad x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$

$R(X) \qquad R(X) = x_{(n)} - x_{(1)} \qquad f(x)$

$[a, b]$ where $f(x) > 0 \qquad x_{(n)} \ge x_{(1)}$

$R(X) \ge 0 \qquad R(X + c) = R(X)$

$R(cX) = |c|R(X)$

PDF 01 / slide 16
$\frac{1}{n} \qquad [0, 1] \qquad x_i' = \frac{x_i - x_{(1)}}{R(X)}$

PDF 01 / slide 17
$x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)} \qquad i = \frac{p}{100} \times n$

($x_{\lfloor i \rfloor}$ and $x_{\lceil i \rceil}$)

PDF 01 / slide 18
$P(X \le x) = 0.25 \qquad P(X \le x) = 0.50$

$P(X \le x) = 0.75$ Probability Density $f(x)$

$\text{IQR} = Q3 - Q1$

$Q1$ (25th percentile) $= -0.67$

$Q2$ / Median (50th) $= 0.00$

$Q3$ (75th percentile) $= 0.67$ Bottom 25%
25% to 50%
50% to 75%
Top 25%

25% of data 25% of data 25% of data 25% of data

PDF 01 / slide 19
$IQR = Q_3 - Q_1$

PDF 01 / slide 20
$\text{IQR} = 1081, \quad \text{Range} = ?$

PDF 01 / slide 23
$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

$n - 1$ (Bessel's Correction) $\quad n - 1$

$s = \sqrt{s^2}$

PDF 01 / slide 24
$z = \frac{x - \bar{x}}{s}$

PDF 01 / slide 25
Measure	Parameter Notation	Statistic Notation
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

PDF 01 / slide 26
$s = 0$

PDF 01 / slide 27
$N = 3{,}226 \qquad \text{Mean} = 3.39\,\text{kg} \qquad \text{SD} = 0.55\,\text{kg}$

PDF 01 / slide 28
$\mu - 3\sigma, \quad \mu - 2\sigma, \quad \mu - \sigma, \quad \mu, \quad \mu + \sigma, \quad \mu + 2\sigma, \quad \mu + 3\sigma$

68%
95%
99.7%

PDF 01 / slide 29
Resting Heart Rate	Deviation from $\bar{x}$	Squared Deviations
50	$50 - 65.5 = -15.5$	240.25
51	$51 - 65.5 = -14.5$	210.25
57	$57 - 65.5 = -8.5$	72.25
61	$61 - 65.5 = -4.5$	20.25
70	$70 - 65.5 = 4.5$	20.25
71	$71 - 65.5 = 5.5$	30.25
80	$80 - 65.5 = 14.5$	210.25
84	$84 - 65.5 = 18.5$	342.25
	Sum of squares:	1146

Sample variance: $s^2 = \frac{1146}{7} \approx 163.71$

Standard deviation, $s = \sqrt{163.71} \approx 12.8$

PDF 01 / slide 30
$0 = \operatorname{Var}(B) < \operatorname{Var}(A) < \operatorname{Var}(D) < \operatorname{Var}(C)$

PDF 02 / slide 02
$y = f(\mathbf{x}) \qquad \operatorname{Var}(\epsilon) = 0$

$T_{\text{Celsius}} = (T_{\text{Fahrenheit}} - 32) \times \frac{5}{9}$

PDF 02 / slide 03
$y = f(\mathbf{x}) + \epsilon \qquad \operatorname{Var}(\epsilon) > 0 \qquad \mathbb{E}[Y \mid \mathbf{x}]$

PDF 02 / slide 04
$F = m\vec{a} \qquad F = m\vec{a}$

PDF 02 / slide 05
$S = \{\textbf{Head}, \textbf{Tail}\}$

PDF 02 / slide 07
Expected Mean (20.0)

PDF 02 / slide 13
$\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\} \qquad S = \{O_i \mid i \in \mathbb{Z}^+\}$

PDF 02 / slide 14
$(D_1, D_2) \qquad \Omega = \{(i, j) \mid i, j \in \{1, 2, 3, 4, 5, 6\}\}$

n_estimators:	$\{100, 200, 300, 400\}$	(4 discrete states)
max_depth:	$\{10, 20, 30\}$	(3 discrete states)
criterion:	$\{\text{gini}, \text{entropy}\}$	(2 discrete states)

PDF 02 / slide 16
$S = \{(x, y, \theta) \mid -\infty < x < \infty, \ 0 < y < d, \ 0 \le \theta \le 2\pi\}$

PDF 02 / slide 17
$x \sim \text{Uniform}(0, d/2) \qquad \theta \sim \text{Uniform}(0, \pi/2)$

$x \in [0, a/2]$ and $y \in [0, b/2]$

$\theta \in [0, \pi/2]$ for $l < \min(a, b)$

PDF 02 / slide 19
$e \in A \qquad e \notin A \qquad C_k \ (x_i \in C_k)$

PDF 02 / slide 20
$A =$ "Exactly one car turns right"

$C =$ "All cars turn in the same direction"

PDF 02 / slide 21
$\varnothing = \{\} \qquad P(\varnothing) = 0$

$S =$ the event that contains all outcomes

PDF 02 / slide 22
$A \cup B \qquad A \cup B = \{e \in S \mid e \in A \text{ or } e \in B\}$

$\{A_i\}_{i \in I}$

$\bigcup_{i \in I} A_i = \{e \in S \mid \exists\, i \in I \text{ such that } e \in A_i\}$

PDF 02 / slide 23
$A, B \subset S \qquad A \cap B = \{e \in S \mid e \in A \text{ and } e \in B\}$

$P(A \cap B)$

PDF 02 / slide 24
$A \cup B$ and $A \cap B$

$A = \{(x, y) : 0 < x < 4, \ 0 < y < 3\}$

$B = \{(x, y) : 2 < x < 4, \ 2 < y < 4\}$

PDF 02 / slide 25
A^c or $\bar{A}$

$A^c = \bar{A} = \{e \mid e \text{ does not belong to } A\}$

$A^c = \bar{A}$

PDF 02 / slide 27
$A =$ "roll an even number"

$B =$ "roll a number less than 4"

$A = \{2, 4, 6\}, \qquad B = \{1, 2, 3\}$

$A \cup B, \quad A \cap B, \quad \bar{A} \quad A \cup B = \qquad A \cap B =$

$\bar{A} =$

PDF 02 / slide 28
$A \cup B = B \cup A, \qquad A \cap B = B \cap A$

$A \cup (B \cup C) = (A \cup B) \cup C$ (Associative law)

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law)

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive law)

$A \cup A = A, \qquad A \cap A = A \qquad \overline{(\bar{A})} = A$

$\overline{A \cup B} = \bar{A} \cap \bar{B}$ (De-Morgan's Law)

$\overline{A \cap B} = \bar{A} \cup \bar{B}$ (De-Morgan's Law)

$A \cup \bar{A} = S, \qquad A \cap \bar{A} = \varnothing$

PDF 02 / slide 29
$A = (A \cap B) \cup (A \cap \bar{B})$

$A = (A \cap B) \cup (A \cap \bar{B})$

PDF 02 / slide 30
$A \cup \phi = A \qquad A \cap \phi = \phi \qquad A \cup S = S$

$A \cap S = A$

PDF 02 / slide 31
$A \cap B = \phi$

PDF 02 / slide 32
$A \subset B$ (equivalently, $B \supset A$)

if $e \in A$, then $e \in B$

$A, B \in \mathcal{P}(\Omega)$, either $A \subsetneq B$ or $B \subsetneq A$

$\Omega = \{1, 2\}, \qquad A = \{1\}, \qquad B = \{2\}$

$A \not\subseteq B$ and $B \not\subseteq A$

PDF 02 / slide 33
If $A \subset B$, then $A \cap B = A \qquad A \cup B = B$

$A \subset B$ iff $\bar{A} \supset \bar{B}$

PDF 02 / slide 34
$A, \quad B, \quad A \cup B, \quad \bar{A} \cap B$

PDF 02 / slide 36
$\bigcup_{i=1}^{k} E_i = E_1 \cup E_2 \cup E_3 \cup \cdots \cup E_k$

$\bigcup_{i=1}^{\infty} E_i = E_1 \cup E_2 \cup E_3 \cup \cdots$

PDF 02 / slide 37
$\bigcap_{i=1}^{k} E_i = E_1 \cap E_2 \cap E_3 \cap \cdots \cap E_k$

$\bigcap_{i=1}^{\infty} E_i = E_1 \cap E_2 \cap E_3 \cap \cdots$

PDF 02 / slide 38
$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c \qquad \left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$

PDF 02 / slide 39
$P(E) \ge 0, \quad \text{for each } E \qquad P(S) = 1$

If $E_1, E_2, \ldots$ are disjoint events, then $P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$

PDF 02 / slide 40
$S = \{o_1, o_2, o_3, \ldots, o_N\}$

each of the outcomes is equally likely: $P(\{o_i\}) = \frac{1}{N}$

$P(E) = \frac{m}{N} = \frac{n(E)}{n(S)} = \frac{n(E)}{N}$

$= \frac{\#\text{ of outcomes in } E}{\text{total } \#\text{ of outcomes}}$

$n(A)$ means the number of elements of A

PDF 02 / slide 41
$A =$ "Exactly one car turns right"

$C =$ "All cars turn in the same direction"

$P(A), \quad P(C)$

PDF 02 / slide 42
$A =$ "roll sum greater than or equal to 10"

$P(A) =$

PDF 02 / slide 43
Finite number of outcomes; all outcomes are equally likely

$P(E) = \frac{n(E)}{n(S)} = \frac{n(E)}{N} = \frac{\#\text{ of outcomes in } E}{\text{total } \#\text{ of outcomes}}$

$n(A) =$ no. of elements of $A \qquad n(E)$ and $n(S)$

PDF 03 / slide 02
$E \subset S \qquad P(E) \ge 0 \qquad P(S) = 1$

$E_1, E_2, \ldots \subset S, \quad E_i \cap E_j = \varnothing$ for all $i \ne j$

$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$

PDF 03 / slide 03
$(\Omega, \mathcal{F}, \mathbb{P}) \qquad \mathbb{P} : \mathcal{F} \to [0, 1]$

PDF 03 / slide 04
$S = \{o_1, o_2, o_3, \ldots, o_N\}$

each of the outcomes is equally likely: $P(\{o_i\}) = \frac{1}{N}$

$P(E) = \frac{m}{N} = \frac{n(E)}{n(S)} = \frac{n(E)}{N}$

$= \frac{\#\text{ of outcomes in } E}{\text{total } \#\text{ of outcomes}}$

$n(A)$ means the number of elements of A

PDF 03 / slide 05
$A =$ "Exactly one car turns right" $= \{LLR, LRL, RLL\}$

$C =$ "All cars turn in the same direction" $= \{LLL, RRR\}$

$P(A), \quad P(C) \qquad P(A) = \frac{3}{8} = 0.375$

$P(C) = \frac{2}{8} = \frac{1}{4} = 0.25$

PDF 03 / slide 06
$A =$ "roll sum greater than or equal to 10"

$P(A) =$

PDF 03 / slide 07
$P(E) \ge 0, \quad \text{for each } E \qquad P(S) = 1$

If A and B are disjoint events, then $P(A \cup B) = P(A) + P(B)$

$P(\bar{A}) = 1 - P(A)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

PDF 03 / slide 08
$W =$ "ace or diamonds" $\qquad P(W)$

$P(W) = P(\text{Ace} \cup \text{Diamonds})$

$= P(\text{Ace}) + P(\text{Diamonds}) - P(\text{Ace of Diamonds})$

$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

PDF 03 / slide 10
$A_i \cap A_j = \phi$ for all $i \ne j$

$n_i = n(A_i) =$ the number of elements in A_i

$A = A_1 \cup A_2 \cup A_3 \cup \cdots$

$N = n(A) =$ the number of elements in A

$= n_1 + n_2 + n_3 + \cdots$

PDF 03 / slide 11
$n_1 =$ number of ways the first step can be performed

number of ways the second step can be

$n_2 =$ performed once the first step has been completed

$N = n_1 n_2 = \frac{\text{number of ways the two operations can}}{\text{be performed in a sequence}}$

PDF 03 / slide 13
$8 \times 7 = 56$

PDF 03 / slide 14
$n_1 =$ number of ways the first operation can be performed

$n_i =$ number of ways the ith operation can be performed once the first $(i-1)$ operations have been completed, $\quad i = 2, 3, \ldots, k$

$N = n_1 n_2 \cdots n_k =$ total number of ways the k operations can be performed in sequence

PDF 03 / slide 17
$(n_1 = n) \qquad (n_2 = n - 1) \qquad (n_k = n - k + 1)$

$(n_n = 1)$

$N = n(n-1)\cdots(n-k+1)\cdots(3)(2)(1) = n!$

$0! = 1$

PDF 03 / slide 18
$\{A, B, C, D\} \qquad N = 4! = 4(3)(2)(1) = 24$

ABCD ABDC ACBD ACDB ADBC ADCB
BACD BADC BCAD BCDA BDAC BDCA
CABD CADB CBAD CBDA CDAB CDBA
DABC DACB DBAC DBCA DCAB DCBA

PDF 03 / slide 20
$k < n \qquad n_1 = n \qquad n_2 = n - 1$

$n_k = n - k + 1$

$N = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$

$P_{n,k} = {}_nP_k = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$

PDF 03 / slide 21
$n = 10$

$P(n, k) = P_{n,k} = P_{10,3} = \frac{10!}{(10-3)!} = \frac{10!}{7!}$

$= 10 \times 9 \times 8 = 720$

PDF 03 / slide 24
$n = 10$

PDF 03 / slide 25
$k \le n \qquad \{A, B, C, D, E\}$

$\{A,B,C\}\ \{A,B,D\}\ \{A,B,E\}\ \{A,C,D\}\ \{A,C,E\}$
$\{A,D,E\}\ \{B,C,D\}\ \{B,C,E\}\ \{B,D,E\}\ \{C,D,E\}$

PDF 03 / slide 26
$\binom{n}{k} \qquad P(n, k) = \frac{n!}{(n-k)!} \qquad k!$

$k! = k \times (k-1) \times \cdots \times 1 \qquad P(n, k) \qquad k!$

$\binom{n}{k} \qquad P(n, k) = \binom{n}{k} \times k! \qquad k!$

$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!}$

PDF 03 / slide 27
$\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$

$= \frac{n(n-1)(n-2)\cdots(n-k+1)}{k(k-1)(k-2)\cdots(1)}$

${}_nC_k = \binom{n}{k} = C(n, k)$ (binomial coefficient)

$(x + y)^n$

PDF 03 / slide 28
$(x+y)^n = {}_nC_0 x^0 y^n + {}_nC_1 x^1 y^{n-1} + {}_nC_2 x^2 y^{n-2} + \cdots + {}_nC_k x^k y^{n-k} + \cdots + {}_nC_n x^n y^0$

$= \binom{n}{0}x^0 y^n + \binom{n}{1}x^1 y^{n-1} + \binom{n}{2}x^2 y^{n-2} + \cdots + \binom{n}{k}x^k y^{n-k} + \cdots + \binom{n}{n}x^n y^0$

PDF 03 / slide 29
$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{k!}, \quad k \in \mathbb{N}, \ \alpha \in \mathbb{R}$

$(1+x)^\alpha = \sum_{k=0}^{\infty}\binom{\alpha}{k}x^k, \quad |x| < 1$

PDF 03 / slide 30
$n(A_1 \cup A_2 \cup A_3 \cup \cdots) = n(A_1) + n(A_2) + n(A_3) + \cdots$ if the sets $A_1, A_2, A_3, \ldots$ are pairwise mutually exclusive (i.e., $A_i \cap A_j = \phi$)

$N = n_1 n_2 = \frac{\text{number of ways the two operations can}}{\text{be performed in a sequence if}}$

$n_1 =$ number of ways the first operation can be performed

number of ways the second operation

$n_2 =$ can be performed once the first has been completed

PDF 03 / slide 31
$N = n_1 n_2 \cdots n_k =$ number of ways k operations can be performed in a sequence if

$n_1 =$ number of ways the first operation can be performed

$n_i =$ number of ways the ith operation can be performed once the first $(i-1)$ operations have been completed, $\quad i = 2, 3, \ldots, k$

PDF 03 / slide 32
Permutations without Replacement (Standard): $P(n, r) = \frac{n!}{(n-r)!}$

Permutations with Replacement: n^r

Permutations of a Multiset: $\frac{n!}{n_1! \times n_2! \cdots \times n_k!}$

PDF 03 / slide 33
Combinations without Replacement (Standard): $C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Combinations with Replacement (Stars and Bars): $\binom{n+r-1}{r}$

PDF 03 / slide 35
$\binom{49}{6} = {}_{49}C_6 = \frac{49!}{6!43!}$

$= \frac{49(48)(47)(46)(45)(44)}{6(5)(4)(3)(2)(1)} = 13{,}983{,}816$

PDF 03 / slide 37
$\binom{49}{6} = 13{,}983{,}816 \qquad \binom{43}{6} = 6{,}096{,}454$

$P(0 \text{ matches}) = \frac{6{,}096{,}454}{13{,}983{,}816} \approx 0.43596$

PDF 03 / slide 38
One winning number: $\binom{6}{1}\binom{43}{5} = 6(962{,}598) = 5{,}775{,}588$

PDF 03 / slide 39
Two winning numbers: $\binom{6}{2} \times \binom{43}{4} = 15 \times 123{,}410 = 1{,}851{,}150$

Three winning numbers: $\binom{6}{3}\binom{43}{3} = 20 \times 12341 = 246{,}820$

Four winning numbers: $\binom{6}{4}\binom{43}{2} = 15 \times 903 = 13{,}545$

PDF 03 / slide 40
Five winning numbers: $\binom{6}{5} \times \binom{43}{1} = 6 \times 43 = 258$

Six winning numbers: $\binom{6}{6} = 1$

PDF 03 / slide 41
$N = \binom{52}{5} = 2{,}598{,}960$

PDF 03 / slide 42
High card $\{x, y, z, u, v\}$ Pair $\{x, x, y, z, u\}$

Two pairs $\{x, x, y, y, z\}$

Three of a kind $\{x, x, x, y, z\}$

Straight $\{x, x+1, x+2, x+3, x+4\}$, 5 cards in sequence, not all the same suit

Flush $\{x, y, z, u, v\}$, not all in sequence but all the same suit

PDF 03 / slide 43
Full House $\{x, x, x, y, y\}$

Four of a Kind $\{x, x, x, x, y\}$

Straight Flush $\{x, x+1, x+2, x+3, x+4\}$

5 cards in sequence but not $\{10, J, Q, K, A\}$, all the same suit

Royal Flush $\{10, J, Q, K, A\}$, all the same suit

PDF 03 / slide 44
VISUALIZING ONE PAIR $\{x, x, y, z, u\}$

Commonness: Approx. 42.26% of all 5-card hands

PDF 03 / slide 45
Pair $\{x, x, y, z, u\} \qquad \binom{13}{1} = 13 \qquad \binom{4}{2} = 6$

$\binom{12}{3} = 220 \qquad \binom{4}{1}^3 = 4^3 = 64$

$13 \times 6 \times 220 \times 64 = 1{,}098{,}240$

PDF 03 / slide 46
Two pair $\{x, x, y, y, z\} \qquad \binom{13}{2} = 78$

$\binom{4}{2} \times \binom{4}{2} = 36 \qquad \binom{11}{1} = 11$

Choose the suit for z − 4

Total # of hands of this type $= 78 \times 36 \times 11 \times 4 = 123{,}552$

PDF 03 / slide 47
Three of a kind $\{x, x, x, y, z\} \qquad \binom{13}{1} = 13$

$\binom{4}{3} = 4$

Choose the denominations $\{y, z\} \qquad \binom{12}{2} = 66$

Choose the suits for $\{y, z\}$ − $4 \times 4 = 16$

Total # of hands of this type $= 13 \times 4 \times 66 \times 16 = 54{,}912$

Full house $\{x, x, x, y, y\} \qquad {}_{13}P_2 = 13(12) = 156$

$\binom{4}{3} = 4 \qquad \binom{4}{2} = 6$

Total # of hands of this type $= 156 \times 4 \times 6 = 3{,}696$

PDF 03 / slide 48
Four of a kind $\{x, x, x, x, y\} \qquad \binom{13}{1} = 13$ ways

$\binom{4}{4} = 1$ way $\qquad \binom{12}{1} = 1$ ways

$\binom{4}{1} = 4$ ways

Total # of hands of this type $= 13 \times 1 \times 12 \times 4 = 624$

Royal Flush $\{10, J, Q, K, A\}$, all the same suit

Total # of hands of this type = 4 (no. of suits)

Straight Flush $\{x, x+1, x+2, x+3, x+4\}$

5 cards in sequence but not $\{10, J, Q, K, A\}$, all the same suit

Total # of hands of this type $= 9 \times 4 = 36$ (no. of suits)

The hand could start with $\{A, 2, 3, 4, 5, 6, 7, 8, 9\}$

PDF 03 / slide 49
Straight $\{x, x+1, x+2, x+3, x+4\}$, 5 cards in sequence, not all the same suit

Total of 10 possibilities $\{A, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$4 \times 4 \times 4 \times 4 \times 4 = 1024$

Total # of hands of this type $= 1024 \times 10 - 36 - 4 = 10200$

PDF 03 / slide 50
Flush $\{x, y, z, u, v\}$, not all in sequence but all the same suit

Choose the denominations $\{x, y, z, u, v\}$

$\binom{13}{5} = 1287$

Total # of hands of this type $= 1287 \times 4 - 36 - 4 = 5108$

PDF 03 / slide 51
	Frequency	Prob.
nothing	1,302,588	0.50119586
pair	1,098,240	0.42256903
two pair	123,552	0.04753902
3 of a kind	54,912	0.02112845
straight	10,200	0.00392465
flush	5,108	0.00196540
full house	3,696	0.00142211
4 of a kind	624	0.00024010
straight flush	36	0.00001385
royal flush	4	0.00000154
Total	2,598,960	1.00000000

High Card / No Pair (50.11%)	50.12%
One Pair (42.26%)	42.26%
Two Pair (4.75%)	4.75%
Three of a Kind (2.11%)	2.11%
All Other Hands (0.76%)	0.76%

PDF 04 / slide 03
$n - 1 \qquad n - 1 \qquad \binom{n}{2} \qquad n - 1 \qquad 1/\binom{n}{2}$

$p = \frac{n-1}{\binom{n}{2}} = \frac{2}{n}$

PDF 04 / slide 04
$3! = 6 \qquad 2/6 = 1/3$

PDF 04 / slide 05
$S = E \cup \overline{E} \quad \& \quad E \cap \overline{E} = \varnothing$

$1 = P[S] = P[E \cup \overline{E}] = P[E] + P[\overline{E}]$

$P[\overline{E}] = 1 - P[E] \qquad \overline{S} = \phi$

$P[\phi] = P[\overline{S}] = 1 - P[S] = 1 - 1 = 0$

PDF 04 / slide 06
$\binom{49}{6} = {}_{49}C_6 = \frac{49!}{6!43!} = \frac{49(48)(47)(46)(45)(44)}{6(5)(4)(3)(2)(1)}$

$= 13{,}983{,}816$

PDF 04 / slide 09
$n \times n \times \cdots \times n = n^m \qquad \binom{m}{k} \qquad m - k$

$n - 1 \qquad m - k \qquad n - 1 \qquad (n-1)^{m-k}$

Favorable Outcomes $= \binom{m}{k}(n-1)^{m-k}$

PDF 04 / slide 10
$P = \frac{\binom{m}{k}(n-1)^{m-k}}{n^m} \qquad n^m \qquad n^k \cdot n^{m-k}$

$P = \binom{m}{k}\left(\frac{1}{n}\right)^k\left(1 - \frac{1}{n}\right)^{m-k}$

PDF 04 / slide 11
$E =$ the event that at least 2 have the same birthday

$\overline{E} =$ the event no pair of individuals have the same birthday

We will find $P(E)$ as $P(E) = 1 - P(\overline{E})$

$n(S) =$ total number of ways of assigning the n individuals birthdays $= 365^n$

$n(\overline{E}) =$ total number of ways of assigning the n distinct birthdays

PDF 04 / slide 12
$P(\overline{E}) = \frac{n(\overline{E})}{n(S)} = \frac{P_{365,n}}{365^n}$

$P[E] = 1 - P[\overline{E}] = 1 - \frac{\binom{365}{n}n!}{365^n}$

$= 1 - \frac{\frac{365!}{n!(365-n)!}n!}{365^n} = 1 - \frac{365!}{(365-n)!365^n}$

$= 1 - \left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right)\cdots\left(\frac{365-n+1}{365}\right)$

PDF 04 / slide 13
$P\left[\begin{array}{c}\text{Two people have same birthday}\\ \text{in a room of } n \text{ people}\end{array}\right] = 1 - \left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right)\cdots\left(\frac{365-n+1}{365}\right)$

n	$P[E]$
10	0.1169
20	0.4114
22	0.4757
23	0.5073
30	0.7063
40	0.8912

$0.5073 \longleftarrow 50\%$

PDF 04 / slide 14
$\frac{n!}{n_1! n_2! \cdots n_r!} \qquad n_1 + \cdots + n_r = n$

PDF 04 / slide 16
$\frac{59!}{4!1!2! \cdots 2!0!} \qquad \frac{1}{\frac{59!}{4!1!2! \cdots 2!0!}} \qquad 1.7 \times 10^{-61}$

PDF 04 / slide 17
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = P(A) + P(B) \quad \text{if } A \cap B = \phi$

PDF 04 / slide 18
$A \cup B$

Only A occurs: $A \cap B^c$ (where B^c is the complement of B)

Only B occurs: $B \cap A^c \qquad$ Both occur: $A \cap B$

$A \cup B$

$P(A \cup B) = P(A \cap B^c) + P(B \cap A^c) + P(A \cap B)$

$A \cap \overline{B} \qquad A \cap B \qquad \overline{A} \cap B$

PDF 04 / slide 19
$A = (A \cap B) \cup (A \cap \overline{B})$

$P(A) = P(A \cap B) + P(A \cap \overline{B})$

$\Rightarrow P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$P(B) = P(A \cap B) + P(\overline{A} \cap B)$

$\Rightarrow P(\overline{A} \cap B) = P(B) - P(A \cap B)$

$P(A \cup B) = P(A \cap \overline{B}) + P(A \cap B) + P(\overline{A} \cap B)$

$= P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B)$

$= P(A) + P(B) - P(A \cap B)$

PDF 04 / slide 21
$P[A \cup B] = P[A] + P[B] - P[A \cap B]$

$A =$ the event that city A is amongst the final 5

$B =$ the event that city B is amongst the final 5

$P[A] = 0.20, \quad P[B] = 0.35, \quad P[A \cap B] = 0.08$

$P[A \cup B]$ "both" $= \cap$, "or" $\equiv \cup$

$P[A \cup B] = P[A] + P[B] - P[A \cap B]$

$= 0.20 + 0.35 - 0.08 = 0.47$

PDF 04 / slide 22
$N = n(S) = \binom{52}{13} = \frac{52(51)(50)(49)\cdots(42)(41)(40)}{13(12)(11)(10)\cdots(3)(2)(1)}$

$= 635{,}013{,}559{,}600$

PDF 04 / slide 24
$A =$ the event that the hand contains exactly 5 spades (8 non-spades)?

$n(A) =$ the number of hands that contain exactly 5 spades (8 non-spades)

$= \binom{13}{5}\binom{39}{8} = (1{,}287)(61{,}523{,}748) = 79{,}181{,}063{,}676$

$\Rightarrow P(A) = \frac{n(A)}{n(S)} = \frac{79{,}181{,}063{,}676}{635{,}013{,}559{,}600} = 0.12469$

PDF 04 / slide 25
$B =$ the event that the hand contains exactly 5 hearts

$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{13}{5}\binom{39}{8}}{\binom{52}{13}}$

$= \frac{79{,}181{,}063{,}676}{635{,}013{,}559{,}600} = 0.1246919$

PDF 04 / slide 26
$A \cap B =$ the event that the hand contains exactly 5 spades and 5 hearts?

$n(A \cap B) = \binom{13}{5}\binom{13}{5}\binom{26}{3}$

$= (1{,}287)(1{,}287)(61{,}523{,}748)$

$= 4{,}306{,}559{,}400$

$P[A \cap B] = \frac{4{,}306{,}559{,}400}{635{,}013{,}559{,}600} = 0.006781838$

PDF 04 / slide 27
$P[A \cup B] = P[A] + P[B] - P[A \cap B]$

$= 0.124691926 + 0.124691926 - 0.006781838$

$= 0.2426$

PDF 04 / slide 28
$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

if $A \cap B = \phi$, $A \cap C = \phi$ and $B \cap C = \phi$, then

$P[A \cup B \cup C] = P[A] + P[B] + P[C]$

PDF 04 / slide 30
$A \cup B \cup C = E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5 \cup E_6 \cup E_7$

$A = E_1 \cup E_2 \cup E_4 \cup E_5$

$B = E_1 \cup E_2 \cup E_3 \cup E_6$

$C = E_1 \cup E_3 \cup E_4 \cup E_7 \qquad A \cap B = E_1 \cup E_2$

$A \cap C = E_1 \cup E_4 \qquad B \cap C = E_1 \cup E_3$

$A \cap B \cap C = E_1$

PDF 04 / slide 31
A = {first die shows 6}, B = {second die shows 6}, C = {third die shows 6}

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

$= \frac{6^2}{6^3} + \frac{6^2}{6^3} + \frac{6^2}{6^3} - \frac{6}{6^3} - \frac{6}{6^3} - \frac{6}{6^3} + \frac{1}{6^3} = \frac{91}{216} = 0.4212963$

PDF 04 / slide 33
$1 \le n \le 2019 \qquad 1442/2019 \qquad 2800/(10^7)$

PDF 04 / slide 34
$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1}P(A_1 \cap \cdots \cap A_n)$

$A_i \cap A_j = \phi \quad \text{for all } i \ne j$

$P\left[\bigcup_{i=1}^{n} A_i\right] = \sum_{i=1}^{n} P[A_i]$

PDF 04 / slide 36
$m \le n \qquad n - m \qquad 1/e$

PDF 05 / slide 01
$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$

PDF 05 / slide 02
$\frac{1}{6} \qquad \frac{1}{3} \qquad P(2) = 1$ of 6 outcomes

$P(2 \mid) = 1$ of 3 even outcomes

PDF 05 / slide 04
$P(A \mid B) \qquad P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

$P(B) \ne 0$

PDF 05 / slide 05
$P(B) \qquad A \cap B \qquad P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$

PDF 05 / slide 06
Total	100%
StreamFlix total	45%
BingeMax total	60%
Only S	20%
Both	25%
Only B	35%
Neither	20%

PDF 05 / slide 07
$P(A \cap B) = P(A) \cdot P(B) \qquad P(A \mid B) = P(A)$

$P(B \mid A) = P(B) \qquad P(A) \ne 0, \quad P(B) \ne 0$

PDF 05 / slide 08
$P(A \cap B) = P(A)P(B)$

$\iff \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{1} = \frac{P(A)}{P(S)}$

$P(A) = 0.5, \quad P(A \cap B) = 0.2, \quad P(B) = 0.4$

$P(A) = 0.5, \quad P(B) = 0.4, \quad P(A \cap B) = 0.2$

$P(A \cap B) = P(A) * P(B) \qquad 0.2 = 0.5 * 0.4$

Total squares = 100

PDF 05 / slide 10
$A_1, A_2, \ldots, A_n; \quad 2 \le k \le n; \quad \{i_1, i_2, \ldots, i_k\}$

$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j})$

$P(A), \quad P(B), \quad P(A \cap B)$

$P(A \cap B) = P(A) * P(B)$

PDF 05 / slide 11
$P(A) = P(B) = P(C) = 0.1 \qquad P(A \cup B)$

$P(A \cup B \cup C)$

PDF 05 / slide 12
$S = \left\{\begin{array}{c}HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT,\\ THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\end{array}\right\}$

prob = 1/16

PDF 05 / slide 13
$C_1 =$ {left ear tag is lost}, $\quad C_2 =$ {right ear tag is lost}

$P(C_1) = P(C_2) = p$

PDF 05 / slide 14
$P[A \cap B] = \begin{cases} P[A]P[B \mid A] & P[A] \ne 0, \\ P[B]P[A \mid B] & P[B] \ne 0. \end{cases}$

$P[A \cap B] = P[A]P[B] \quad (A, B \text{ independent})$

$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \times P(A_2 \mid A_1) \times P(A_3 \mid A_1 \cap A_2) \times \cdots \times P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$

PDF 05 / slide 17
$B_1 =$ first ball is black, $\quad R_2 =$ second ball is red

$P(B_1 \cap R_2)$

$P(B_1 \cap R_2) = P(B_1) \cdot P(R_2 \mid B_1)$

PDF 05 / slide 18
$P(B_1) \qquad r + b \qquad P(B_1) = \frac{b}{r+b}$

$P(R_2 \mid B_1) \qquad P(R_2 \mid B_1) = \frac{r}{r+b-1}$

$P(B_1 \cap R_2) = \left(\frac{b}{r+b}\right) \cdot \left(\frac{r}{r+b-1}\right) = \frac{br}{(r+b)(r+b-1)}$

PDF 05 / slide 19
$P(R_2) \qquad R_1 \cap R_2, \quad B_1 \cap R_2$

$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2) \qquad R_1 \cap R_2$

$P(R_1 \cap R_2) = P(R_1) \cdot P(R_2 \mid R_1) = P(R_1)$

$= \left(\frac{r}{r+b}\right)\left(\frac{r-1}{r+b-1}\right)$

$= \frac{r(r-1)}{(r+b)(r+b-1)}$

$B_1 \cap R_2$

$P(B_1 \cap R_2) = P(B_1) \cdot P(R_2 \mid B_1) = \frac{br}{(r+b)(r+b-1)}$

$P(R_2) = \frac{r(r-1)}{(r+b)(r+b-1)} + \frac{br}{(r+b)(r+b-1)}$

$= \frac{r[(r-1)+b]}{(r+b)(r+b-1)} = \frac{r(r+b-1)}{(r+b)(r+b-1)}$

$(r + b - 1) \qquad P(R_2) = \frac{r}{r+b}$

PDF 05 / slide 21
$P(B) = P(A_1)P(B \mid A_1) + \ldots + P(A_k)P(B \mid A_k)$

PDF 05 / slide 23
$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2)}$

$P(A_2 \mid B) = \frac{P(B \mid A_2)P(A_2)}{P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2)}$

$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$

$P(A), \quad P(B \mid A), \quad P(A \mid B)$

PDF 05 / slide 24
$P(A_1 \mid B) = \frac{P(A_1 \cap B)}{P(B)}$

PDF 05 / slide 25
$P(A_1 \cap B) = P(B \mid A_1)P(A_1)$

$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{P(B)}$

PDF 05 / slide 26
$P(B) = P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2)$

$P(B)$

$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2)}$

PDF 05 / slide 27
Sensitivity $= P(\text{test is positive} \mid \text{disease}) = 0.96$

Specificity $= P(\text{test is negative} \mid \text{no disease}) = 1 - 0.02 = 0.98$

PDF 05 / slide 28
$A =$ person has the disease, $\quad \bar{A} =$ person has no disease

$P[A] = 0.001, \quad P[\bar{A}] = 1 - 0.001 = 0.999$

$B =$ test is positive

$P[B \mid A] = 0.96, \quad P[B \mid \bar{A}] = 0.02 \qquad P[A \mid B]$

PDF 05 / slide 29
$P(A \mid B) = \frac{P(A \cap B)}{P(A)P(B \mid A) + P(\bar{A})P(B \mid \bar{A})}$

$= \frac{0.001 \times 0.95}{0.001 \times 0.95 + 0.999 \times 0.02} = 0.045$

$p = 0.001$

$P(\text{disease}) = 0.001, \quad P(\text{no disease}) = 1 - 0.001 = 0.999$

$p = 0.045$

$P(\text{disease} \mid T+) = 0.045, \quad P(\text{no disease} \mid T+) = 0.955$

PDF 05 / slide 30
$P\left(\bigcup_i A_i\right) = \sum_i P(A_i) = 1$

$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_{j=1}^{\infty} P(B \mid A_j)P(A_j)} \qquad P(A_i)$

$P(A_i \mid B)$

PDF 05 / slide 32
$P(A) = \frac{1}{2} \qquad P(B) = \frac{1}{2}$

$P(R_1 R_2 \mid A) = \frac{12}{20} \times \frac{11}{19} = \frac{132}{380}$

$P(R_1 R_2 \mid B) = \frac{4}{20} \times \frac{3}{19} = \frac{12}{380}$

PDF 05 / slide 33
$P(R_1 R_2)$

$P(R_1 R_2) = P(R_1 R_2 \mid A)P(A) + P(R_1 R_2 \mid B)P(B)$

$P(R_1 R_2) = \left(\frac{132}{380}\right)\left(\frac{1}{2}\right) + \left(\frac{12}{380}\right)\left(\frac{1}{2}\right) = \frac{144}{760} = \frac{18}{95}$

$P(A \mid R_1 R_2) = \frac{P(R_1 R_2 \mid A)P(A)}{P(R_1 R_2)}$

$= \frac{\frac{132}{760}}{\frac{144}{760}} = \frac{132}{144} = \frac{11}{12} \approx 91.67\%$

$P(B \mid R_1 R_2) = 1 - \frac{11}{12} = \frac{1}{12} \approx 8.33\%$

PDF 05 / slide 34
$P(R_3 \mid R_1 R_2) \qquad \frac{11}{12} \qquad P(R_3 \mid A, R_1 R_2) = \frac{10}{18}$

$\frac{1}{12} \qquad P(R_3 \mid B, R_1 R_2) = \frac{2}{18}$

PDF 05 / slide 35
$P(R_3 \mid R_1 R_2) = P(R_3 \mid A, R_1 R_2) \cdot P(A \mid R_1 R_2) + P(R_3 \mid B, R_1 R_2) \cdot P(B \mid R_1 R_2)$

$P(R_3 \mid R_1 R_2) = \left(\frac{10}{18}\right)\left(\frac{11}{12}\right) + \left(\frac{2}{18}\right)\left(\frac{1}{12}\right)$

$P(R_3 \mid R_1 R_2) = \frac{110}{216} + \frac{2}{216} = \frac{112}{216} = \frac{14}{27} \approx 51.85\%$

PDF 05 / slide 37
$\begin{array}{c|c} +1 & p \\ -1 & 1-p \end{array} \qquad p < 0.5, \quad p = 0.45$

$p > 0.5, \quad p = 0.55 \qquad p = 0.5 \qquad y = 0 \qquad y = 0$

$0 < p < 1 \qquad 1 - p \qquad k - i \qquad k - i$

PDF 05 / slide 39
$q = 1 - p \qquad P_0 = 0 \qquad P_k = 1$

PDF 05 / slide 40
$1 \le i \le k - 1 \qquad i + 1 \qquad q = 1 - p, \quad i - 1$

$P_i = pP_{i+1} + qP_{i-1}$

$p + q = 1, \quad P_i = (p + q)P_i$

$(p + q)P_i = pP_{i+1} + qP_{i-1}$

$p(P_{i+1} - P_i) = q(P_i - P_{i-1})$

$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$

PDF 05 / slide 41
$P_j - P_{j-1}, \quad \frac{q}{p}$

$P_2 - P_1 = \left(\frac{q}{p}\right)(P_1 - P_0) = \left(\frac{q}{p}\right)P_1$

$P_3 - P_2 = \left(\frac{q}{p}\right)(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1$

$\vdots$

$P_i - P_{i-1} = \left(\frac{q}{p}\right)^{i-1} P_1$

$j = 1$

$P_i - P_0 = \sum_{j=1}^{i}(P_j - P_{j-1}) = P_1\sum_{j=0}^{i-1}\left(\frac{q}{p}\right)^j$

$P_0 = 0 \qquad P_i = P_1\sum_{j=0}^{i-1}\left(\frac{q}{p}\right)^j$

PDF 05 / slide 42
$q = 1 - p \qquad P_i = pP_{i+1} + qP_{i-1}$

$pP_{i+1} - P_i + qP_{i-1} = 0 \qquad P_i = r^i, \quad r^i$

$pr^{i+1} - r^i + qr^{i-1} = 0 \qquad r \ne 0, \quad r^{i-1}$

$pr^2 - r + q = 0 \qquad p + q = 1, \quad (p + q)$

$pr^2 - (p + q)r + q = 0 \qquad (pr - q)(r - 1) = 0$

$r_1 = 1, \quad r_2 = \frac{q}{p}$

$p \ne 0.5, \quad q \ne p, \quad r_1 = 1, \quad r_2 = \frac{q}{p}$

$P_i = A(r_1)^i + B(r_2)^i$

$P_i = A(1)^i + B\left(\frac{q}{p}\right)^i = A + B\left(\frac{q}{p}\right)^i$

PDF 05 / slide 43
$p \ne \frac{1}{2}, \quad p \ne q \qquad \sum_{j=0}^{i-1} x^j = \frac{1 - x^i}{1 - x}, \quad x = \frac{q}{p}$

$P_i = P_1\frac{1 - (q/p)^i}{1 - q/p} \qquad P_k = 1$

$1 = P_1\frac{1 - (q/p)^k}{1 - q/p} \Longrightarrow P_1 = \frac{1 - q/p}{1 - (q/p)^k}$

$P_i = \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \left(\frac{q}{p}\right)^k} = \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^k}$

PDF 05 / slide 44
$p = \frac{1}{2}, \quad p = q \qquad \frac{q}{p} = 1$

$P_i = P_1\sum_{j=0}^{i-1} 1 = i \cdot P_1$

$P_k = 1: \quad 1 = k \cdot P_1 \Longrightarrow P_1 = \frac{1}{k} \qquad P_i = \frac{i}{k}$

$P_i = \begin{cases} \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^k} & \text{if } p \ne \frac{1}{2}, \\ \frac{i}{k} & \text{if } p = \frac{1}{2}. \end{cases}$

PDF 06 / slide 03
$X =$ the total number of ads clicked during the session

S	X
NNN	0
CNN	1
NCN	1
NNC	1
NCC	2
CCN	2
CNC	2
CCC	3

PDF 06 / slide 06
$p(x) \qquad p(x) = P(X = x) \qquad x \in \mathbb{R}$

PDF 06 / slide 07
(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	3	4	5	6	7
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	4	5	6	7	8
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	5	6	7	8	9
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	6	7	8	9	10
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	7	8	9	10	11
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	8	9	10	11	12

PDF 06 / slide 08
$p(2) = P[X = 2] = P[\{(1, 1)\}] = \frac{1}{36}$

$p(3) = P[X = 3] = P[\{(1, 2), (2, 1)\}] = \frac{2}{36}$

$p(4) = P[X = 4] = P[\{(1, 3), (2, 2), (3, 1)\}] = \frac{3}{36}$

$p(5) = \frac{4}{36}, \quad p(6) = \frac{5}{36}, \quad p(7) = \frac{6}{36}, \quad p(8) = \frac{5}{36}, \quad p(9) = \frac{4}{36}$

$p(10) = \frac{3}{36}, \quad p(11) = \frac{2}{36}, \quad p(12) = \frac{1}{36}$

$p(x) = 0$ for all other x

PDF 06 / slide 10
$F_X(x) \qquad F_X(x) = P(X \le x)$

$0 \le F_X(x) \le 1$ for all $x \in \mathbb{R}$

$\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$

If $a \le b$, then $F_X(a) \le F_X(b)$

$\lim_{t \to x^+} F_X(t) = F_X(x)$

PDF 06 / slide 11
$F_X(x): \quad \frac{0}{36}, \frac{3}{36}, \frac{6}{36}, \frac{9}{36}, \frac{12}{36}, \frac{15}{36}, \frac{18}{36}, \frac{21}{36}, \frac{24}{36}, \frac{27}{36}, \frac{30}{36}, \frac{33}{36}, \frac{36}{36}$

$F(x)$ is a step function

PDF 06 / slide 12
$P[E] = \frac{\text{area of } E}{\text{Area of } S} = \text{area of } E$

PDF 06 / slide 13
$F(x) = P[X \le x] = P\left[\left\{\begin{array}{c}\text{set of all points within a}\\ \text{dist } x \text{ from lower left corner}\end{array}\right\}\right]$

$0 \le x \le 1 \qquad 1 \le x \le \sqrt{2} \qquad \sqrt{2} \le x$

PDF 06 / slide 14
$F(x) = P[X \le x] = \begin{cases} 0 & x \le 0 \\ \frac{\pi x^2}{4} & 0 \le x \le 1 \\ \text{Area } A & 1 \le x \le \sqrt{2} \\ 1 & \sqrt{2} \le x \end{cases}$

$0 \le x \le 1 \qquad 1 \le x \le \sqrt{2} \qquad \sqrt{2} \le x$

PDF 06 / slide 15
$1 \le x \le \sqrt{2} \qquad \sqrt{x^2 - 1} \qquad \sqrt{x^2 - 1}$

$\theta = \frac{\pi}{2} - 2\phi \qquad \tan(\phi) = \sqrt{x^2 - 1}$

$\phi = \tan^{-1}\left(\sqrt{x^2 - 1}\right)$

$A = 2\left(\frac{1\sqrt{x^2 - 1}}{2}\right) + \frac{\theta}{2\pi}(\pi x^2)$

$= \sqrt{x^2 - 1} + \frac{\theta}{2}x^2$

$= \sqrt{x^2 - 1} + \left(\frac{\pi}{4} - \phi\right)x^2$

$= \sqrt{x^2 - 1} + \left(\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right)x^2$

PDF 06 / slide 16
$F(x) = P[X \le x] = \begin{cases} 0 & x \le 0 \\ \frac{\pi x^2}{4} & 0 \le x \le 1 \\ \sqrt{x^2 - 1} + \left(\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right)x^2 & 1 \le x \le \sqrt{2} \\ 1 & \sqrt{2} \le x \end{cases}$

PDF 06 / slide 17
$-\infty < x < \infty \qquad f(x) \ge 0, \quad \forall x \in \mathbb{R}$

$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(a \le X \le b) = \int_a^b f(x)\,dx$

PDF 06 / slide 18
$f(x) \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$

$P[a \le X \le b] = \int_a^b f(x)\,dx$

PDF 06 / slide 19
$F(x) \qquad F(x) = P[X \le x] = \int_{-\infty}^{x} f(t)\,dt$

$F(x)$

PDF 06 / slide 20
$F(x) = P[X \le x] = \int_{-\infty}^{x} f(t)\,dt$

$F'(x) = \frac{dF(x)}{dx} = f(x)$

PDF 06 / slide 21
$F(x) = P(X \le x)$

$F(-\infty) = 0 \quad \& \quad F(\infty) = 1$

$F(x) = \int_{-\infty}^{x} f(u)\,du \qquad f(x) = F'(x)$

$f(x)$ slope

PDF 06 / slide 22
$F_X(x) = \frac{1}{12}(x^2 + x^3), \quad x \in [0, 2] \qquad$ Find $f(x)$

PDF 06 / slide 23
$F(x) = P[X \le x] = \begin{cases} 0 & x \le 0 \\ \frac{\pi x^2}{4} & 0 \le x \le 1 \\ \sqrt{x^2 - 1} + \left(\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right)x^2 & 1 \le x \le \sqrt{2} \\ 1 & \sqrt{2} \le x \end{cases}$

$f(x) = F'(x) = \begin{cases} 0, & x \le 0 \text{ or } x \ge \sqrt{2} \\ \frac{\pi x}{2}, & 0 \le x \le 1 \\ \frac{d}{dx}\left\{\sqrt{x^2 - 1} + \left[\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right]x^2\right\}, & 1 \le x \le \sqrt{2} \end{cases}$

PDF 06 / slide 24
$\frac{d}{dx}\left\{\sqrt{x^2 - 1} + \left[\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right]x^2\right\}$

$= \frac{1}{2}(x^2 - 1)^{-\frac{1}{2}}(2x) + \frac{\pi}{2}x - 2x\tan^{-1}\left(\sqrt{x^2 - 1}\right)$

$- x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2 - 1}\right)$

$= \frac{\pi}{2}x + \frac{x}{(x^2 - 1)^{\frac{1}{2}}} - 2x\tan^{-1}\left(\sqrt{x^2 - 1}\right)$

$- x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2 - 1}\right)$

PDF 06 / slide 25
$\frac{d}{du}\tan^{-1}u = \frac{1}{1 + u^2}$

$\Rightarrow \frac{d}{dx}\tan^{-1}\left(\sqrt{x^2 - 1}\right) = \frac{1}{1 + (x^2 - 1)}\left(\frac{1}{2}\right)(x^2 - 1)^{-\frac{1}{2}}(2x)$

$\Rightarrow -x^2\frac{d}{dx}\tan^{-1}\left(\sqrt{x^2 - 1}\right) = -\frac{x}{(x^2 - 1)^{\frac{1}{2}}}$

$\Rightarrow \frac{d}{dx}\left\{\sqrt{x^2 - 1} + \left[\frac{\pi}{4} - \tan^{-1}\left(\sqrt{x^2 - 1}\right)\right]x^2\right\}$

$= \frac{\pi}{2}x - 2x\tan^{-1}\left(\sqrt{x^2 - 1}\right)$

$f(x) = F'(x) = \begin{cases} 0, & \text{if } x \le 0 \text{ or } x \ge \sqrt{2} \\ \frac{\pi x}{2}, & \text{if } x \in [0, 1] \\ \frac{\pi x}{2} - 2x\arctan\left(\sqrt{x^2 - 1}\right), & \text{if } x \in [1, \sqrt{2}] \end{cases}$

PDF 06 / slide 26
$x = 0 \qquad P(X = 0) = 0.4 \qquad 0 < x \le 2$

$f(x) = 0.3x \qquad F(x) \qquad X \ge 1$

Answer for b): 0.45

PDF 06 / slide 27
$p(x) \qquad 0 \le p(x) \le 1 \qquad \sum p(x) = 1$

$P(a \le X \le b) = \sum_{a \le x \le b} p(x)$

$P(a \le X \le b)$ might be different than $P(a < X < b)$

PDF 06 / slide 28
x	3	5	8	10
$P(x)$	0.3	0.2	0.3	?

$P(X = 10) = 1 - (0.3 + 0.2 + 0.3) = 0.2$

$P(X \ge 5) = p(5) + p(8) + p(10) = 0.7$

$P(X > 5) = p(8) + p(10) = 0.5$

$P(4 < X \le 8) = p(5) + p(8) = 0.5$

PDF 06 / slide 29
$F(x) \qquad F(x) = P[X \le x]$

$F(-\infty) = 0$ and $F(\infty) = 1$

$\{X \le -\infty\} = \phi$ and $\{X \le \infty\} = S$

$F(-\infty) = 0$ and $F(\infty) = 1$

PDF 06 / slide 30
If $x_1 < x_2$ then $F(x_1) \le F(x_2)$

If $x_1 < x_2$ then $\{X \le x_2\} = \{X \le x_1\} \cup \{x_1 < X \le x_2\}$

$P[X \le x_2] = P[X \le x_1] + P[x_1 < X \le x_2]$

$F(x_2) = F(x_1) + P[x_1 < X \le x_2]$

$P[x_1 < X \le x_2] \ge 0$ then $F(x_2) \ge F(x_1)$

$F(b) - F(a) = P[a < X \le b] \qquad a < b$

$F(b) = F(a) + P[a < X \le b]$

$F(b) - F(a) = P[a < X \le b]$

PDF 06 / slide 31
$p(x) = P[X = x] = F(x) - F(x-)$

$F(x-) = \lim_{u \to x-} F(u)$

If $p(x) = 0$ for all x, then $F(x)$ is continuous

$F(x-) = \lim_{u \to x-} F(u) = F(x+) = \lim_{u \to x+} F(u)$

$p(x) = 0$ implies that $F(x+) = F(x) = F(x-)$

PDF 06 / slide 32
$F(x) = P[X \le x] = \sum_{u \le x} p(u)$

$F(-\infty) = 0$ and $F(\infty) = 1$

$p(x) = F(x) - F(x-) =$ jump in $F(x)$ at x

PDF 06 / slide 33
$x = 1, 2, 3, \ldots$

$p(x) = c \cdot x \cdot \left(\frac{1}{3}\right)^x, \quad \text{for } x \in \{1, 2, 3, \ldots\}$

$p(x) \qquad P(X > 2)$

PDF 06 / slide 34
$X =$ sum of the two numbers showing

PDF 06 / slide 35
$X =$ sum of the two numbers showing

X	2	3	$\cdots$	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\cdots$	$\frac{1}{36}$

PDF 07 / slide 01
$f(x) \ge 0 \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$

$P(a \le X \le b) = \int_a^b f(x)\,dx$

$f(x) = F'(x), \quad F$ is the cdf

$P(a \le X \le b) = P(a < X \le b) = P(a \le X < b) = P(a < X < b)$

PDF 07 / slide 02
$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P[a < X < b] = \int_a^b f(x)\,dx$

PDF 07 / slide 03
$F(x) = P(X \le x), \quad \forall x \in \mathbb{R} \qquad \lim_{x \to \infty} F(x) = 1$

$\lim_{x \to -\infty} F(x) = 0 \qquad P(a < X \le b) = F(b) - F(a)$

$F'(x) = f(x) \qquad F_X(x) = 0 \quad \text{for } x < x_1$

$P_X(x_1), P_X(x_2), P_X(x_k) \qquad \lim_{x \to \infty} F_X(x) = 1$

$P(X = 0), \quad P(X = 1)$

PDF 07 / slide 04
$f(x) = cx^n, \quad 0 < x < 1 \qquad$ Find c

Find $P(X > 0.05)$

PDF 07 / slide 05
$f(x) = cx^n, \quad 0 < x < 1 \qquad$ Find c

Find $P(X > 0.05)$

$\int_0^1 f(x)\,dx = 1 \iff \int_0^1 cx^n\,dx = 1$

$\iff c\int_0^1 x^n\,dx = 1 \iff c\left.\frac{x^{n+1}}{n+1}\right|_{x=0}^{1} = 1$

$\iff c\left(\frac{1^{n+1}}{n+1} - \frac{0^{n+1}}{n+1}\right) = 1 \iff \frac{c}{n+1} = 1 \iff c = n + 1$

$P(x > 0.05) = \int_{0.05}^{1}(n+1)x^n\,dx = x^{n+1}\big|_{0.05}^{1} = 1 - (0.05)^{n+1}$

PDF 07 / slide 06
Find $P(X > 2 \mid X \le 4)$

PDF 07 / slide 07
$F(x) = P(X \le x) = \begin{cases} 0, & x < -2 \\ \frac{1}{2} + \frac{3}{32}\left(4x - \frac{x^3}{3}\right), & -2 \le x < 2 \\ 1, & 2 \le x \end{cases}$

$P(X < 0) \qquad P(-1 < X < 1) \qquad P(0.5 < X)$

$f(x)$ median of X

PDF 07 / slide 09
$p(x, y) = P[X = x, Y = y] \qquad 0 \le p(x, y) \le 1$

$\sum_x\sum_y p(x, y) = 1$

$P[(X, Y) \in A] = \sum_{(x,y) \in A}\sum p(x, y)$

PDF 07 / slide 10
$N = 25 \qquad n = 7$

$X =$ the number of images with Minor Errors in the sample

$Y =$ the number of Completely Mislabeled images in the sample

Derive $p(x, y)$

PDF 07 / slide 11
$p(x, y) = P(X = x, Y = y)$

$p(x, y) = \frac{\binom{8}{x}\binom{5}{y}\binom{12}{7-x-y}}{\binom{25}{7}} \qquad X + Y \le n$

$X: 0, 1, 2, 3, 4, 5, 6, 7 \quad (n = 7)$

$Y: 0, 1, 2, 3, 4, 5 \qquad X + Y \le 7$

PDF 07 / slide 12
$p(x, y) = \frac{\binom{8}{x}\binom{5}{y}\binom{12}{7-x-y}}{\binom{25}{7}}$

Y\X	X = 0	X = 1	X = 2	X = 3	X = 4	X = 5	X = 6	X = 7
Y = 0	0.0016	0.0154	0.0461	0.0577	0.0320	0.0077	0.0007	< 0.0001
Y = 1	0.0096	0.0659	0.1442	0.1281	0.0481	0.0070	0.0003	0.0000
Y = 2	0.0165	0.0824	0.1281	0.0769	0.0175	0.0012	0.0000	0.0000
Y = 3	0.0103	0.0366	0.0384	0.0140	0.0015	0.0000	0.0000	0.0000
Y = 4	0.0023	0.0055	0.0035	0.0006	0.0000	0.0000	0.0000	0.0000
Y = 5	0.0001	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000

PDF 07 / slide 14
$n = 5 \qquad X =$ the number of times a six appears

$Y =$ the number of times a five appears

$p(x, y) = P(X = x, Y = y) \qquad X: 0, 1, 2, 3, 4, 5$

$Y: 0, 1, 2, 3, 4, 5 \qquad X + Y \le 5$

PDF 07 / slide 15
F5FF6, $\quad F \in \{1, 2, 3, 4\} \qquad \left(\frac{1}{6}\right)^1\left(\frac{1}{6}\right)^1\left(\frac{4}{6}\right)^3$

$\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{4}{6}\right)^{5-x-y}$

$x =$ the number of sixes in the sequence

$y =$ the number of fives in the sequence

PDF 07 / slide 16
$p(x, y) = P(X = x, Y = y) = K\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{4}{6}\right)^{5-x-y}$

$K =$ number of sequences of length 5 containing x sixes and y fives

$= \binom{5}{x}\binom{5-x}{y}\binom{5-x-y}{5-x-y}$

$= \frac{5!}{x!(5-x)!}\frac{(5-x)!}{y!(5-x-y)!} = \frac{5!}{x!y!(5-x-y)!}$

PDF 07 / slide 17
$p(x, y) = P[X = x, Y = y] = \frac{5!}{x!y!(5-x-y)!}\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{4}{6}\right)^{5-x-y}$

$x + y \le 5$

PDF 07 / slide 18
$p(x, y) = \frac{5!}{x!y!(5-x-y)!}\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{4}{6}\right)^{5-x-y}$

	0	1	2	3	4	5
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001
1	0.1646	0.1646	0.0617	0.0103	0.0006	0
2	0.0823	0.0617	0.0154	0.0013	0	0
3	0.0206	0.0103	0.0013	0	0	0
4	0.0026	0.0006	0	0	0	0
5	0.0001	0	0	0	0	0

PDF 07 / slide 19
$n = 5 \qquad X =$ the number of times a six appears

$Y =$ the number of times a five appears

$P(X > Y)$

PDF 07 / slide 20
$p(x, y) = \frac{5!}{x!y!(5-x-y)!}\left(\frac{1}{6}\right)^x\left(\frac{1}{6}\right)^y\left(\frac{4}{6}\right)^{5-x-y}$

	0	1	2	3	4	5
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001
1	0.1646	0.1646	0.0617	0.0103	0.0006	0
2	0.0823	0.0617	0.0154	0.0013	0	0
3	0.0206	0.0103	0.0013	0	0	0
4	0.0026	0.0006	0	0	0	0
5	0.0001	0	0	0	0	0

$P[X > Y] = \sum_{x > y}\sum p(x, y) = 0.3441$

PDF 07 / slide 21
$p(x, y) = P(X = x, Y = y)$

$p_X(x) = P(X = x) = \sum_y p(x, y)$

$p_Y(y) = P(Y = y) = \sum_x p(x, y)$

PDF 07 / slide 22
$y_1, y_2, y_3, \ldots$ possible values of Y

$p_X(x) = P[X = x]$

$= P[\{X = x, Y = y_1\} \cup \{X = x, Y = y_2\} \cup \cdots]$

$= P[X = x, Y = y_1] + P[X = x, Y = y_2] + \cdots$

$= p(x, y_1) + p(x, y_2) + \cdots$

$= \sum_j p(x, y_j) = \sum_y p(x, y)$

$p_Y(y) = \sum_x p(x, y)$

PDF 07 / slide 23
$n = 5 \qquad X =$ the number of times a six appears

$Y =$ the number of times a five appears

	0	1	2	3	4	5	$p_X(x)$
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001	0.4019
1	0.1646	0.1646	0.0617	0.0103	0.0006	0	0.4019
2	0.0823	0.0617	0.0154	0.0013	0	0	0.1608
3	0.0206	0.0103	0.0013	0	0	0	0.0322
4	0.0026	0.0006	0	0	0	0	0.0032
5	0.0001	0	0	0	0	0	0.0001
$p_Y(y)$	0.4019	0.4019	0.1608	0.0322	0.0032	0.0001	

PDF 07 / slide 24
$X \in \{0, 1\}: \quad 0 = \text{Retained}, \ 1 = \text{Churned}$

$Y \in \{1, 2, 3\}: \quad 1 = \text{Low}, \ 2 = \text{Medium}, \ 3 = \text{High}$

$P(X, Y)$

X\Y	1	2	3
0	0.05	0.25	0.4
1	0.15	0.1	0.05

PDF 07 / slide 25
X\Y	1	2	3	
0	0.05	0.25	0.4	0.7
1	0.15	0.1	0.05	0.3
	0.2	0.35	0.45	

X	0	1
$p(x)$	0.7	0.3

Y	1	2	3
$p(y)$	0.2	0.35	0.45

PDF 07 / slide 27
$p(x, y) = P[X = x, Y = y]$

$p_{X|Y}(x \mid y) = P[X = x \mid Y = y] = \frac{p(x, y)}{p_Y(y)}, \quad Y = y$

$p_{Y|X}(y \mid x) = P[Y = y \mid X = x] = \frac{p(x, y)}{p_X(x)}, \quad X = x$

PDF 07 / slide 29
$n = 5 \qquad X =$ the number of times a six appears

$Y =$ the number of times a five appears

x\y	0	1	2	3	4	5	$p_X(x)$
0	0.1317	0.1646	0.0823	0.0206	0.0026	0.0001	0.4019
1	0.1646	0.1646	0.0617	0.0103	0.0006	0	0.4019
2	0.0823	0.0617	0.0154	0.0013	0	0	0.1608
3	0.0206	0.0103	0.0013	0	0	0	0.0322
4	0.0026	0.0006	0	0	0	0	0.0032
5	0.0001	0	0	0	0	0	0.0001
$p_Y(y)$	0.4019	0.4019	0.1608	0.0322	0.0032	0.0001	

PDF 07 / slide 30
$p_{X|Y}(x \mid y) = P[X = x \mid Y = y], \quad y = 0, 1, \ldots, 5$

x\y	0	1	2	3	4	5
0	0.3277	0.4096	0.5120	0.6400	0.8000	1.0000
1	0.4096	0.4096	0.3840	0.3200	0.2000	0.0000
2	0.2048	0.1536	0.0960	0.0400	0.0000	0.0000
3	0.0512	0.0256	0.0080	0.0000	0.0000	0.0000
4	0.0064	0.0016	0.0000	0.0000	0.0000	0.0000
5	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000

PDF 07 / slide 31
$p_{Y|X}(y \mid x) = P[Y = y \mid X = x], \quad x = 0, 1, \ldots, 5$

x\y	0	1	2	3	4	5
0	0.3277	0.4096	0.2048	0.0512	0.0064	0.0003
1	0.4096	0.4096	0.1536	0.0256	0.0016	0.0000
2	0.5120	0.3840	0.0960	0.0080	0.0000	0.0000
3	0.6400	0.3200	0.0400	0.0000	0.0000	0.0000
4	0.8000	0.2000	0.0000	0.0000	0.0000	0.0000
5	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000

PDF 07 / slide 32
Group	Event Occurred	Event Did Not Occur	Total
Exposed	a	b	$a + b$
Unexposed	c	d	$c + d$

Relative Risk $(RR) = \frac{\text{Risk in Exposed Group}}{\text{Risk in Unexposed Group}} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}}$

PDF 07 / slide 33
$X \in \{0, 1\}: \quad 0 = \text{Retained}, \ 1 = \text{Churned}$

$Y \in \{1, 2, 3\}: \quad 1 = \text{Low}, \ 2 = \text{Medium}, \ 3 = \text{High}$

$P(X, Y)$

X\Y	1	2	3
0	0.05	0.25	0.4
1	0.15	0.1	0.05

Churn Status (X) given Low Engagement Tier $(Y = 1)$

PDF 07 / slide 34
$X \in \{1, 2, 3\}: \quad 1 = \text{Basic}, \ 2 = \text{Pro}, \ 3 = \text{Enterprise}$

$Y \in \{0, 1, 2\}$: number of API threshold limit breaches in a given hour

$p(x, y) = P(X = x, Y = y)$

$p(x, y) = k \cdot \frac{x^y}{y!} \cdot 2^{-x}, \quad x \in \{1, 2, 3\}, \ y \in \{0, 1, 2\}$

Find $k \qquad P(X = x), \quad P(Y = y)$

$\hat{X}$ when observing $Y = 2$

PDF 07 / slide 35
$\sum_{x=1}^{3}\sum_{y=0}^{2} p(x, y) = 1 \qquad f(x, y) = \frac{x^y}{y!}2^{-x}$

$f(1, 0) = \frac{1^0}{0!}2^{-1} = \frac{1}{2} = \frac{8}{16}$

$f(1, 1) = \frac{1^1}{1!}2^{-1} = \frac{1}{2} = \frac{8}{16}$

$f(1, 2) = \frac{1^2}{2!}2^{-1} = \frac{1}{4} = \frac{4}{16}$

Subtotal for $x = 1$: $\frac{20}{16}$

$f(2, 0) = \frac{2^0}{0!}2^{-2} = \frac{1}{4} = \frac{4}{16}$

$f(2, 1) = \frac{2^1}{1!}2^{-2} = \frac{2}{4} = \frac{8}{16}$

$f(2, 2) = \frac{2^2}{2!}2^{-2} = \frac{2}{4} = \frac{8}{16}$

Subtotal for $x = 2$: $\frac{20}{16}$

$f(3, 0) = \frac{3^0}{0!}2^{-3} = \frac{1}{8} = \frac{2}{16}$

$f(3, 1) = \frac{3^1}{1!}2^{-3} = \frac{3}{8} = \frac{6}{16}$

$f(3, 2) = \frac{3^2}{2!}2^{-3} = \frac{9}{16}$

Subtotal for $x = 3$: $\frac{17}{16}$

$\sum_{x=1}^{3}\sum_{y=0}^{2} f(x, y) = \frac{20 + 20 + 17}{16} = \frac{57}{16}$

$k \cdot \frac{57}{16} = 1 \qquad k = \frac{16}{57}$

PDF 07 / slide 36
$k = \frac{16}{57}$

X\Y	Y = 0	Y = 1	Y = 2	P(X = x)
X = 1	$\frac{8}{57}$	$\frac{8}{57}$	$\frac{4}{57}$	$\frac{20}{57}$
X = 2	$\frac{4}{57}$	$\frac{8}{57}$	$\frac{8}{57}$	$\frac{20}{57}$
X = 3	$\frac{2}{57}$	$\frac{6}{57}$	$\frac{9}{57}$	$\frac{17}{57}$
P(Y = y)	$\frac{14}{57}$	$\frac{22}{57}$	$\frac{21}{57}$	1.0

PDF 08 / slide 02
$0 \le f(x, y) \qquad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\,dx\,dy = 1$

$P[(X, Y) \in A] = \iint_A f(x, y)\,dx\,dy$

$z = f(x, y)$

PDF 08 / slide 03
$(x, y) \in R \qquad \iint_R f(x, y)\,dx\,dy$

PDF 08 / slide 04
$A = \{(x, y) \mid a \le x \le b, \ c \le y \le d\}$

$\iint_A f(x, y)\,dx\,dy = \int_c^d\left[\int_a^b f(x, y)\,dx\right]dy = \int_a^b\left[\int_c^d f(x, y)\,dy\right]dx$

PDF 08 / slide 05
$\iint_A f(x, y)\,dx\,dy = \int_c^d\int_a^b f(x, y)\,dx\,dy$

$= \int_c^d\left[\int_a^b f(x, y)\,dx\right]dy$

$G(y) = \int_a^b f(x, y)\,dx$

$\int_c^d\int_a^b f(x, y)\,dx\,dy = \int_c^d G(y)\,dy$

PDF 08 / slide 06
$\iint_A f(x, y)\,dx\,dy = \int_a^b\int_c^d f(x, y)\,dy\,dx$

$= \int_a^b\left[\int_c^d f(x, y)\,dy\right]dx$

$H(x) = \int_c^d f(x, y)\,dy$

$\int_a^b\int_c^d f(x, y)\,dy\,dx = \int_a^b H(x)\,dx$

PDF 08 / slide 07
$\int_0^1\int_0^1(x^2y + xy^3)\,dx\,dy = \int_0^1\int_0^1(x^2y + xy^3)\,dy\,dx$

$\int_0^1\int_0^1(x^2y + xy^3)\,dx\,dy = \int_0^1\left[\int_0^1(x^2y + xy^3)\,dx\right]dy$

$= \int_0^1\left[\frac{x^3}{3}y + \frac{x^2}{2}y^3\right]_{x=0}^{x=1}dy$

$= \int_0^1\left(\frac{1}{3}y + \frac{1}{2}y^3\right)dy = \frac{1}{3}\frac{y^2}{2} + \frac{1}{2}\frac{y^4}{4}\Big|_{y=0}^{y=1}$

$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$

PDF 08 / slide 08
$\int_0^1\int_0^1(x^2y + xy^3)\,dy\,dx = \int_0^1\left[\int_0^1(x^2y + xy^3)\,dy\right]dx$

$= \int_0^1\left[\frac{y^2}{2}x^2 + \frac{y^4}{4}x\right]_{y=0}^{y=1}dx$

$= \int_0^1\left(\frac{1}{2}x^2 + \frac{1}{4}x\right)dx = \frac{1}{2}\frac{x^3}{3} + \frac{1}{4}\frac{x^2}{2}\Big|_{x=0}^{x=1}$

$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$

PDF 08 / slide 09
$\int_0^2\int_0^1 xy^2\,dy\,dx$

PDF 08 / slide 11
$A = \{(x, y) \mid a \le x \le b, \ g_1(x) \le y \le g_2(x)\}$

$y = g_2(x), \quad y = g_1(x) \qquad y = g_2(x), \quad y = g_1(x)$

$y = g_2(x), \quad y = g_1(x)$

$\iint_A f(x, y)\,dx\,dy = \int_a^b\left[\int_{g_1(x)}^{g_2(x)} f(x, y)\,dy\right]dx$

PDF 08 / slide 12
$A = \{(x, y) \mid a(y) \le x \le b(y), \ c \le y \le d\}$

$\iint_A f(x, y)\,dx\,dy = \int_c^d\left[\int_{a(y)}^{b(y)} f(x, y)\,dx\right]dy$

$y = (x + 2)^2 \qquad x = y - \left(\frac{1}{16}\right)y^3$

$(0, 4), \quad (-2, 0), \quad (0, -4)$

PDF 08 / slide 13
$f(x, y) = x^2y + xy^3$

$A = \{(x, y) \mid x + y \le 1, \ 0 \le x, \ 0 \le y\}$

$x + y = 1 \qquad (0, 1), \quad (1, 0)$

$y \ge 0, \quad x + y \le 1, \quad x \ge 0$

PDF 08 / slide 14
$x + y = 1 \qquad (0, 1), \quad (0, y), \quad (1 - y, y), \quad (1, 0)$

$\iint_A(x^2y + xy^3)\,dx\,dy = \int_0^1\int_0^{1-y}(x^2y + xy^3)\,dx\,dy$

PDF 08 / slide 15
$\int_0^1\left[\int_0^{1-y}(x^2y + xy^3)\,dx\right]dy = \int_0^1\left[\frac{x^3}{3}y + \frac{x^2}{2}y^3\right]_{x=0}^{x=1-y}dy$

$= \int_0^1\left[\frac{(1-y)^3}{3}y + \frac{(1-y)^2}{2}y^3\right]dy$

$= \int_0^1\left[\frac{y - 3y^2 + 3y^3 - y^4}{3} + \frac{y^3 - 2y^4 + y^5}{2}\right]dy$

$= \frac{\frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5}}{3} + \frac{\frac{1}{4} - \frac{2}{5} + \frac{1}{6}}{2}$

$= \frac{1}{6} - \frac{1}{3} + \frac{1}{4} - \frac{1}{15} + \frac{1}{8} - \frac{1}{5} + \frac{1}{12}$

$= \frac{20 - 40 + 30 - 8 + 15 - 24 + 10}{120} = \frac{3}{120} = \frac{1}{40}$

PDF 08 / slide 16
$x + y = 1 \qquad (0, 1), \quad (1, 0), \quad (x, 0), \quad (x, 1 - x)$

$\iint_A(x^2y + xy^3)\,dy\,dx = \int_0^1\int_0^{1-x}(x^2y + xy^3)\,dy\,dx$

$= \int_0^1\left[\int_0^{1-x}(x^2y + xy^3)\,dy\right]dx$

PDF 08 / slide 17
$\int_0^1\left[\int_0^{1-x}(x^2y + xy^3)\,dy\right]dx = \int_0^1\left[x^2\frac{y^2}{2} + x\frac{y^4}{4}\right]_{y=0}^{y=1-x}dx$

$= \int_0^1\left[x^2\frac{(1-x)^2}{2} + x\frac{(1-x)^4}{4}\right]dx$

$= \int_0^1\left[\frac{x^2 - 2x^3 + x^4}{2} + \frac{x - 4x^2 + 6x^3 - 4x^4 + x^5}{4}\right]dx$

$= \int_0^1\frac{x - 2x^2 + 2x^3 - 2x^4 + x^5}{4}\,dx = \frac{x^2}{8} - \frac{x^3}{6} + \frac{x^4}{8} - \frac{x^5}{10} + \frac{x^6}{24}\Big|_0^1$

$= \frac{1}{8} - \frac{1}{6} + \frac{1}{8} - \frac{1}{10} + \frac{1}{24} = \frac{15 - 20 + 15 - 12 + 5}{120} = \frac{3}{120} = \frac{1}{40}$

PDF 08 / slide 18
$f(x, y) = 2 - 3x + xy, \quad (0, 0), \ (1, 0), \ (1, 3)$

$\iint_D(3 - 6xy)\,dx\,dy \qquad y = x^2, \quad y = -x^2$

Answer: b) 36

PDF 08 / slide 20
$f(x, y) = ce^{x^2}, \quad 0 \le y \le x \le 1$

$\int_0^1\int_y^1 ce^{x^2}\,dx\,dy = c\int_0^1\int_y^1 e^{x^2}\,dx\,dy = 1$

$\int e^{x^2}\,dx \qquad y \le x \le 1, \quad 0 \le y \le 1$

$(0, 0), \quad (1, 0), \quad (1, 1) \qquad 0 \le y \le x, \quad 0 \le x \le 1$

$\int_0^1\int_0^x e^{x^2}\,dy\,dx$

PDF 08 / slide 21
$f(x, y) = \frac{1}{\pi r^2}, \quad x^2 + y^2 \le r^2 \qquad \frac{r}{2}$

$P(A) = P\left((x, y) : x^2 + y^2 \le \frac{r^2}{4}\right) = \iint_A\frac{1}{r^2}\,dx\,dy$

$= \frac{\text{Area of } A}{\text{Area of } S} = \frac{\pi\left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}$

PDF 08 / slide 22
$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\,dy$

$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\,dx$

PDF 08 / slide 23
$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)}, \quad X = x$

$f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)}, \quad Y = y$

PDF 08 / slide 24
$f(x, y) = (6/5)(x + y^2), \quad 0 < x < 1, \ 0 < y < 1$

$P(X < \frac{1}{4}, Y < \frac{1}{4}) = \int_0^{1/4}\int_0^{1/4}\frac{6}{5}(x + y^2)\,dx\,dy$

$= \frac{6}{5}\int_0^{1/4}\left(\frac{x^2}{2} + y^2x\right)\Big|_{x=0}^{1/4}dy = \cdots$

$f_X(x), \quad f_Y(y)$

$P(\frac{1}{4} < Y < \frac{3}{4}) = \int_{1/4}^{3/4} f_Y(y)\,dy = \frac{1}{5}\int_{1/4}^{3/4}(3 + 6y^2)\,dy = \frac{37}{80}$

PDF 08 / slide 25
$f(x, y) = (6/5)(x + y^2), \quad 0 < x < 1, \ 0 < y < 1$

$\int_0^1\int_0^1\frac{6}{5}(x + y^2)\,dx\,dy \overset{?}{=} 1$

$\frac{6}{5}\int_0^1\left(\frac{x^2}{2} + y^2x\right)\Big|_{x=0}^{1}dy = \frac{6}{5}\int_0^1\left(\frac{1}{2} + y^2\right)dy$

$= \frac{6}{5}\left(\frac{y}{2} + \frac{y^3}{3}\right)\Big|_0^1 = \frac{6}{5}\left(\frac{1}{2} + \frac{1}{3}\right) = 1$

PDF 08 / slide 26
$f(x, y) = (6/5)(x + y^2), \quad 0 < x < 1, \ 0 < y < 1$

$f_X(x), \quad f_Y(y)$

$f(y) = \int_0^1 f(x, y)\,dx = \int_0^1\frac{6}{5}(x + y^2)\,dx$

$= \frac{6}{5}\left(\frac{x^2}{2} + y^2x\right)\Big|_{x=0}^{1} = \frac{6}{5}\left(\frac{1}{2} + y^2\right)$

$= \frac{3}{5} + \frac{6y^2}{5} = \frac{3 + 6y^2}{5}$

$\Longrightarrow f_Y(y) = \frac{3 + 6y^2}{5}, \quad y \in (0, 1)$

PDF 08 / slide 27
$f(x, y) = cx + 1, \quad x, y > 0, \ x + y < 1 \qquad c$

$f_X(x), \ f_Y(y) \qquad f(x \mid y), \ f(y \mid x)$

PDF 08 / slide 28
$f(x, y) = cx + 1, \quad x, y > 0, \ x + y < 1 \qquad$ Find c

$x + y = 1 \qquad \int_0^1\int_0^{1-x}(cx + 1)\,dy\,dx$

$= \int_0^1(cx + 1)y\big|_{y=0}^{1-x}dx$

$= \int_0^1(cx + 1)(1 - x)\,dx = \int_0^1(cx + 1 - cx^2 - x)\,dx$

$= -\frac{cx^3}{3} + \frac{(c-1)x^2}{2} + x\Big|_0^1$

$-\frac{c}{3} + \frac{c-1}{2} + 1 = 1 \qquad c = 3$

PDF 08 / slide 30
$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}}e^{-\frac{1}{2}Q(x_1, x_2)}$

$Q(x_1, x_2) = \frac{\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1 - \mu_1}{\sigma_1}\right)\left(\frac{x_2 - \mu_2}{\sigma_2}\right) + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2}{1 - \rho^2}$

PDF 08 / slide 31
$\sigma_1 = \sigma_2, \quad \rho = 0 \qquad \sigma_1 = \sigma_2, \quad \rho = 0.90$

$\sigma_1 = \sigma_2, \quad \rho = -0.90$

PDF 08 / slide 32
$f(x_1, x_2) = \frac{1}{(2\pi)\sigma_1\sigma_2\sqrt{1 - \rho^2}}e^{-\frac{1}{2}Q(x_1, x_2)}$

$Q(x_1, x_2) = \frac{\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1 - \mu_1}{\sigma_1}\right)\left(\frac{x_2 - \mu_2}{\sigma_2}\right) + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2}{1 - \rho^2}$

$Q(x_1, x_2)$ is constant

PDF 08 / slide 33
$\sigma_1 = \sigma_2, \quad \rho = 0 \qquad \sigma_1 = \sigma_2, \quad \rho = 0.9$

$\sigma_1 = \sigma_2, \quad \rho = -0.9 \qquad \sigma_1 = \sigma_2, \quad \rho = 0$

$\sigma_1 = \sigma_2, \quad \rho = 0.9 \qquad \sigma_1 = \sigma_2, \quad \rho = -0.9$

$\sigma_1 = \sigma_2, \quad \rho = 0 \qquad \sigma_1 = \sigma_2, \quad \rho = 0.9$

$\sigma_1 = \sigma_2, \quad \rho = -0.9$

PDF 08 / slide 34
marginal distribution of x_i is Normal with mean μ_i and standard deviation σ_i, 1, 2

x_2: Normal with mean μ_2 and standard deviation σ_2

x_1: Normal with mean μ_1 and standard deviation σ_1

PDF 08 / slide 35
X_i given $X_j, \quad i, j = 1, 2$ is normal

$\mu_{i|j} = \mu_i + \rho\frac{\sigma_i}{\sigma_j}(x_j - \mu_j) \qquad \sigma_{i|j} = \sigma_i\sqrt{1 - \rho^2}$

PDF 08 / slide 36
$f_{2|1}(x_2 \mid x_1) = \frac{f(x_1, x_2)}{f_1(x_1)}$

$= \frac{\frac{e^{-\frac{1}{2}Q(x_1, x_2)}}{(2\pi)\sigma_1\sigma_2\sqrt{1 - \rho^2}}}{\frac{1}{\sqrt{2\pi}\sigma_1}e^{-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2}}$

$= \frac{e^{-\frac{1}{2}Q(x_1, x_2) - \frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2}}{\sqrt{2\pi}\sigma_1\sqrt{1 - \rho^2}}$

$= \frac{e^{-\frac{1}{2}\left[\left(\frac{x_2 - a}{b}\right)^2 + c\right] - \frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2}}{\sqrt{2\pi}\sigma_1\sqrt{1 - \rho^2}}$

PDF 08 / slide 37
$b = \sigma_1\sqrt{1 - \rho^2} \qquad a = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu)$

$c = \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2$

$f_{1|2}(x_1 \mid x_2) = \frac{1}{\sqrt{2\pi}b}e^{-\frac{1}{2}\left(\frac{x_1 - a}{b}\right)^2}$

conditional distribution of x_2 given x_1 is Normal

$a = \mu_{1|2} = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$

$b = \sigma_{1|2} = \sigma_1\sqrt{1 - \rho^2}$

PDF 08 / slide 38
$k \ge 2$

PDF 08 / slide 39
$f(x_1, \ldots, x_k) = f(\mathbf{x}) = \frac{1}{(2\pi)^{k/2}\sqrt{\det(\boldsymbol{\Sigma})}}e^{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}$

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_k \end{bmatrix}$

$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1k} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1k} & \sigma_{2k} & \cdots & \sigma_{kk} \end{bmatrix}$

PDF 08 / slide 40
$f(x_1, x_2, \ldots, x_n) \qquad f(x_1, \ldots, x_n) \ge 0$

$\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1, \ldots, x_n)\,dx_1\cdots dx_n = 1$

$P[(X_1, \ldots, X_n) \in A] = \int\cdots\int_A f(x_1, \ldots, x_n)\,dx_1\cdots dx_n$

PDF 08 / slide 41
$f(x_1, x_2, \ldots, x_k)$

$f(x_1, \ldots, x_k) = f_1(x_1)f_2(x_2)\cdots f_k(x_k)$

PDF 08 / slide 42
$p(x_1, \ldots, x_n) = \frac{n!}{x_1!x_2!\cdots x_k!}p_1^{x_1}p_2^{x_2}\cdots p_k^{x_k}$

PDF 08 / slide 43
$\frac{n!}{x_1!x_2!\cdots x_k!} = \binom{n}{x_1 \ x_2 \ \cdots \ x_k}$

$\binom{n}{x_1}\binom{n - x_1}{x_2}\binom{n - x_1 - x_2}{x_3}\cdots\binom{x_k}{x_k}$

$= \left(\frac{n!}{x_1!(n - x_1)!}\right)\left(\frac{(n - x_1)!}{x_2!(n - x_1 - x_2)!}\right)\left(\frac{(n - x_1 - x_2)!}{x_3!(n - x_1 - x_2 - x_3)!}\right)\cdots$

$= \frac{n!}{x_1!x_2!\cdots x_k!}$

PDF 08 / slide 44
$p(x_1, \ldots, x_k) = \frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$

$= \binom{n}{x_1 \ x_2 \ \ldots \ x_k}p_1^{x_1}p_2^{x_2}\cdots p_k^{x_k}$

PDF 08 / slide 45
$p_1 = 0.30, \quad p_2 = 0.50, \quad p_3 = 0.20 \qquad n = 4$

$X =$ number that result in a complete cure

$Y =$ number that result in just reduced pain

$Z =$ number that result in no change

$P[X + Y \ge Z]$

$p(x, y, z) = \frac{4!}{x!y!z!}(0.30)^x(0.50)^y(0.20)^z, \quad x + y + z = 4$

PDF 08 / slide 46
$p(x, y, z):$

x	y	$z = 0$	$z = 1$	$z = 2$	$z = 3$	$z = 4$
0	0	0	0	0	0	0.0016
0	1	0	0	0	0.0160	0
0	2	0	0	0.0600	0	0
0	3	0	0.1000	0	0	0
0	4	0.0625	0	0	0	0
1	0	0	0	0	0.0096	0
1	1	0	0	0.0720	0	0
1	2	0	0.1800	0	0	0
1	3	0.1500	0	0	0	0
1	4	0	0	0	0	0
2	0	0	0	0.0216	0	0
2	1	0	0.1080	0	0	0
2	2	0.1350	0	0	0	0
2	3	0	0	0	0	0
2	4	0	0	0	0	0
3	0	0	0.0216	0	0	0
3	1	0.0540	0	0	0	0
3	2	0	0	0	0	0
3	3	0	0	0	0	0
3	4	0	0	0	0	0
4	0	0.0081	0	0	0	0
4	1	0	0	0	0	0
4	2	0	0	0	0	0
4	3	0	0	0	0	0
4	4	0	0	0	0	0

PDF 08 / slide 47
$P[X + Y \ge Z] = 0.9728$

x	y	$z = 0$	$z = 1$	$z = 2$	$z = 3$	$z = 4$
0	0	0	0	0	0	0.0016
0	1	0	0	0	0.0160	0
0	2	0	0	0.0600	0	0
0	3	0	0.1000	0	0	0
0	4	0.0625	0	0	0	0
1	0	0	0	0	0.0096	0
1	1	0	0	0.0720	0	0
1	2	0	0.1800	0	0	0
1	3	0.1500	0	0	0	0
1	4	0	0	0	0	0
2	0	0	0	0.0216	0	0
2	1	0	0.1080	0	0	0
2	2	0.1350	0	0	0	0
2	3	0	0	0	0	0
2	4	0	0	0	0	0
3	0	0	0.0216	0	0	0
3	1	0.0540	0	0	0	0
3	2	0	0	0	0	0
3	3	0	0	0	0	0
3	4	0	0	0	0	0
4	0	0.0081	0	0	0	0
4	1	0	0	0	0	0
4	2	0	0	0	0	0
4	3	0	0	0	0	0
4	4	0	0	0	0	0

PDF 08 / slide 48
$f(x_1, x_2, \ldots, x_q, x_{q+1}, \ldots, x_n)$

$f_{12\ldots q}(x_1, \ldots, x_q) = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1, \ldots, x_n)\,dx_{q+1}\cdots dx_n$

PDF 08 / slide 49
$f(x_1, x_2, \ldots, x_q, x_{q+1}, \ldots, x_k)$

$X_{q+1} = x_{q+1}, \ldots, X_k = x_k$

$f_{1\ldots q|q+1\ldots k}(x_1, \ldots, x_q \mid x_{q+1}, \ldots, x_k) = \frac{f(x_1, \ldots, x_k)}{f_{q+1\ldots k}(x_{q+1}, \ldots, x_k)}$

PDF 09 / slide 02
$E(X) = \sum_x xp(x) = \sum_i x_ip(x_i)$

$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$

PDF 09 / slide 04
X (Outages)	0	1	2	3	4
P(X)	0.15	0.35	0.25	0.15	0.10

$C(X) = 500X^2 + 1000 \qquad$ Find $E[X]$, $E[C(X)]$

PDF 09 / slide 05
X (Outages)	0	1	2	3	4
P(X)	0.15	0.35	0.25	0.15	0.10

Find $E(X) \qquad E[X] = \sum x \cdot P(x)$

$E[X] = (0)(0.15) + (1)(0.35) + (2)(0.25) + (3)(0.15) + (4)(0.10)$

$E[X] = 0 + 0.35 + 0.50 + 0.45 + 0.40$

$E[X] = 1.70$ outages

PDF 09 / slide 07
$f(x) = \frac{3}{2}(1 - x^2), \quad 0 \le x \le 1$

Find $E(X)$, $F(x)$

PDF 09 / slide 08
$f(x) = \frac{3}{2}(1 - x^2), \quad 0 \le x \le 1 \qquad$ Find $E(X)$

$E(X) = \int_0^1 xf(x)\,dx = \int_0^1 x\frac{3}{2}(1 - x^2)\,dx$

$= \frac{3}{2}\int_0^1(x - x^3)\,dx = \frac{3}{2}\left(\left[\frac{x^2}{2}\right]_0^1 - \left[\frac{x^4}{4}\right]_0^1\right)$

$= \frac{3}{2}\left(\frac{1}{2} - 0 - \left(\frac{1}{4} - 0\right)\right) = \frac{3}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{3}{8}$

PDF 09 / slide 09
$f(x) = a + bx^2, \quad 0 \le x \le 1 \qquad E(X) = \frac{3}{5}$

PDF 09 / slide 10
$f(x) = a + bx^2, \quad 0 \le x \le 1 \qquad E(X) = \frac{3}{5}$

$\int f(x)\,dx = 1, \qquad \int xf(x)\,dx = \frac{3}{5}$

$\int_0^1(a + bx^2)\,dx = 1$

$a[x]_0^1 + \left[\frac{bx^3}{3}\right]_{x=0}^{1} = 1$

$a + \frac{b}{3} = 1$

$3a + b = 3$

$\int_0^1 x(a + bx^2)\,dx = \frac{3}{5}$

$a\left[\frac{x^2}{2}\right]_0^1 + b\left[\frac{x^4}{4}\right]_0^1 = \frac{3}{5}$

$\frac{a}{2} + \frac{b}{4} = \frac{3}{5} \quad (\times 20)$

$10a + 5b = 12$

PDF 09 / slide 11
$3a + b = 3 \Rightarrow b = 3 - 3a$

$10a + 5b = 12$

$10a + 5(3 - 3a) = 12$

$10a + 15 - 15a = 12$

$-5a = -3 \Rightarrow a = \frac{3}{5}$

$\Rightarrow b = 3 - 3\left(\frac{3}{5}\right) = \frac{6}{5}$

PDF 09 / slide 12
the expected value of X, $E(X)$

PDF 09 / slide 13
$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b, \\ 0, & x < a, \ x > b, \end{cases}$

$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx = \int_a^b x\frac{1}{b-a}\,dx$

$= \left[\frac{1}{b-a}\frac{x^2}{2}\right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$

$\frac{1}{b-a}, \quad b - a \qquad E(x) = \frac{a+b}{2}$

PDF 09 / slide 15
$E[g(X)] = \sum_x g(x)p(x) = \sum_i g(x_i)p(x_i)$

$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)\,dx$

PDF 09 / slide 16
X (Outages)	0	1	2	3	4
P(X)	0.15	0.35	0.25	0.15	0.10

Find $E[C(X)], \quad C(X) = 500X^2 + 1000$

PDF 09 / slide 17
X (Outages)	0	1	2	3	4
P(X)	0.15	0.35	0.25	0.15	0.10

$E[aX^2 + b] = aE[X^2] + b$

$E[X^2] = \sum x^2 \cdot P(x)$

$= (0^2)(0.15) + (1^2)(0.35) + (2^2)(0.25) + (3^2)(0.15) + (4^2)(0.10)$

$= 0 + 0.35 + 1.00 + 1.35 + 1.60 = 4.30$

$E[C(X)] = E[500X^2 + 1000] = 500E[X^2] + 1000$

$E[C(X)] = 500(4.30) + 1000$

$E[C(X)] = 2150 + 1000 = \$3{,}150$

PDF 09 / slide 18
$f(x) = 3x^2, \quad x \in (0, 1) \qquad$ Find $E\left(\frac{1}{X}\right)$

PDF 09 / slide 19
$f(x) = x^2\left(2x + \frac{3}{2}\right), \quad x \in (0, 1)$

Find $E\left(\frac{2}{X} + 3\right)$

PDF 09 / slide 21
$\mu_k = E(X^k) = \begin{cases} \sum_x x^k p(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} x^k f(x)\,dx, & \text{if } X \text{ is continuous,} \end{cases}$

$\mu = \mu_1 = E(X)$

PDF 09 / slide 22
$\mu_k^0 = E[(X - \mu)^k] \qquad \mu = \mu_1 = E(X)$

PDF 09 / slide 23
$\mu_1^0 = E(X - \mu) = 0$

$\mu_2^0 = E[(X - \mu)^2] =$ 2nd central moment

$\mu_2^0 = E[(X - \mu)^2]$ is called variance of X

PDF 09 / slide 24
$\sigma = \sqrt{\mu_2^0} = \sqrt{\mathbb{E}[(X - \mu)^2]}$

$\operatorname{var}(X) = \mu_2^0 = E[(X - \mu)^2] = \sigma^2 \qquad z = \frac{x - \mu}{\sigma}$

PDF 09 / slide 25
$\mu_3^0 = \mathbb{E}[(X - \mu)^3] \qquad \mu_3^0 > 0 \qquad \mu_3^0 < 0$

PDF 09 / slide 26
$\mu_3^0 = \mathbb{E}[(X - \mu)^3]$

$\gamma_1 = \frac{\mu_3^0}{\sigma^3} = \frac{\mathbb{E}[(X - \mu)^3]}{(\mathbb{E}[(X - \mu)^2])^{3/2}}$

PDF 09 / slide 27
$\mu_3^0 = 0, \quad \gamma_1 = 0 \qquad \mu_3^0 > 0, \quad \gamma_1 > 0$

$\mu_3^0 < 0, \quad \gamma_1 < 0$

PDF 09 / slide 28
$\mu_4^0 = \mathbb{E}[(X - \mu)^4]$

$\operatorname{Kurt}[X] = \frac{\mu_4^0}{\sigma^4} - 3 = \frac{\mathbb{E}[(X - \mu)^4]}{(\mathbb{E}[(X - \mu)^2])^2} - 3$

PDF 09 / slide 31
X	1	2	3
$p(x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

Find $E(X)$, standard deviation σ, μ_3, μ_3^0

PDF 09 / slide 32
X	1	2	3
$p(x)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

$E(X) = 1 * \left(\frac{1}{2}\right) + 2 * \left(\frac{1}{3}\right) + 3 * \left(\frac{1}{6}\right) = \frac{5}{3} = \mu$

$\operatorname{Var}(X) = \sum(x - \mu)^2 \cdot p(x)$

$= \left(1 - \frac{5}{3}\right)^2\left(\frac{1}{2}\right) + \left(2 - \frac{5}{3}\right)^2\left(\frac{1}{3}\right) + \left(3 - \frac{5}{3}\right)^2\left(\frac{1}{6}\right) = \frac{5}{9}$

$\Rightarrow \sigma = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$

$\mu_3 = \sum x^3 \cdot p(x) = 1^3\left(\frac{1}{2}\right) + 2^3\left(\frac{1}{3}\right) + 3^3\left(\frac{1}{6}\right) = \frac{23}{3}$

d) $\cdots \frac{7}{27}$

PDF 09 / slide 33
$f(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & x < 0, \ x > 1, \end{cases}$

$\mu_k = \int_{-\infty}^{\infty} x^k f(x)\,dx = \int_0^1 x^k\,dx = \left[\frac{x^{k+1}}{k+1}\right]_0^1 = \frac{1}{k+1}$

PDF 09 / slide 34
$\mu_k^0 = \int_{-\infty}^{\infty}(x - \mu)^k f(x)\,dx = \int_0^1\left(x - \frac{1}{2}\right)^k 1\,dx$

$w = x - \frac{1}{2}$

$\mu_k^0 = \int_{-1/2}^{1/2} w^k\,dw = \left[\frac{w^{k+1}}{k+1}\right]_{-1/2}^{1/2} = \frac{(1/2)^{k+1} - (-1/2)^{k+1}}{k+1}$

$= \frac{1 - (-1)^{k+1}}{2^{k+1}(k+1)} = \begin{cases} \frac{1}{2^k(k+1)} & \text{if } k \text{ even,} \\ 0 & \text{if } k \text{ odd.} \end{cases}$

PDF 09 / slide 35
$\mu_2^0 = \frac{1}{2^2(3)} = \frac{1}{12}, \quad \mu_4^0 = \frac{1}{2^4(5)} = \frac{1}{80}$

$\operatorname{var}(X) = \mu_2^0 = \frac{1}{12}$

$\sigma = \sqrt{\operatorname{var}(X)} = \sqrt{\frac{1}{12}} \qquad \gamma_1 = \frac{\mu_3^0}{\sigma^3} = 0$

$\gamma_2 = \frac{\mu_4^0}{\sigma^4} - 3 = \frac{1/80}{(1/12)^2} - 3 = -\frac{6}{5}$

PDF 09 / slide 36
$E[c] = c, \quad c$ is constant $\qquad g(X) = c$

$E[g(X)] = E[c] = \int_{-\infty}^{\infty} cf(x)\,dx = c\int_{-\infty}^{\infty} f(x)\,dx = c$

PDF 09 / slide 37
$E[aX + b] = aE[X] + b, \quad a, b$ are constants

$g(X) = aX + b$

$E[aX + b] = \int_{-\infty}^{\infty}(ax + b)f(x)\,dx$

$= a\int_{-\infty}^{\infty} xf(x)\,dx + b\int_{-\infty}^{\infty} f(x)\,dx = aE[X] + b$

PDF 09 / slide 38
$\operatorname{var}(X) = \mu_2^0 = E[(X - \mu)^2] = E[X^2] - (E[X])^2 = \mu_2 - \mu^2$

$\operatorname{var}(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty}(x - \mu)^2 f(x)\,dx$

$= \int_{-\infty}^{\infty}(x^2 - 2\mu x + \mu^2)f(x)\,dx$

$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - 2\mu\int_{-\infty}^{\infty} xf(x)\,dx + \mu^2\int_{-\infty}^{\infty} f(x)\,dx$

$= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2$

PDF 09 / slide 39
$\operatorname{var}(aX + b) = a^2\operatorname{var}(X)$

$\mu_{aX+b} = E[aX + b] = aE[X] + b = a\mu + b$

$\operatorname{var}(aX + b) = E[(aX + b - a\mu - b)^2]$

$= E[(a(X - \mu))^2]$

$= E[a^2(X - \mu)^2] = a^2\operatorname{var}(X)$

PDF 09 / slide 40
$f(x) = x^2\left(2x + \frac{3}{2}\right), \quad x \in (0, 1)$

Find $\operatorname{Var}\left(\frac{2}{X} + 3\right)$

$\operatorname{Var}\left(\frac{2}{X} + 3\right) = 2^2\operatorname{Var}\left(\frac{1}{X}\right)$

$\operatorname{Var}\left(\frac{1}{X}\right) = E\left(\left(\frac{1}{X}\right)^2\right) - \left[E\left(\frac{1}{X}\right)\right]^2$

$= \int_0^1\frac{1}{x^2}x^2\left(2x + \frac{3}{2}\right)dx - \cdots$

$= \int_0^1\left(2x + \frac{3}{2}\right)dx - \cdots$

PDF 09 / slide 42
$E[g(X)] = \begin{cases} \sum_x g(x)p(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} g(x)f(x)\,dx, & \text{if } X \text{ is continuous,} \end{cases}$

$m_X(t) = E[e^{tX}] = \begin{cases} \sum_x e^{tx}p(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} e^{tx}f(x)\,dx, & \text{if } X \text{ is continuous,} \end{cases}$

PDF 09 / slide 43
$X \sim \text{Bin}(n, p)$

$p(x) = \binom{n}{x}p^x(1 - p)^{n-x}, \quad x = 0, 1, 2, \ldots, n$

$M_X(t) = (pe^t + q)^n$

PDF 09 / slide 44
$X \sim \text{Bin}(n, p) \qquad M_X(t) = E[e^{tX}]$

$M_X(t) = \sum_{x=0}^{n} e^{tx}\binom{n}{x}p^xq^{n-x}$

$M_X(t) = \sum_{x=0}^{n}\binom{n}{x}(pe^t)^xq^{n-x}$

$(a + b)^n = \sum_{k=0}^{n}\binom{n}{k}a^kb^{n-k}, \quad a = pe^t, \quad b = q$

$M_X(t) = (pe^t + q)^n$

PDF 09 / slide 45
$X \sim \text{Bin}(n, p)$

$p(x) = \binom{n}{x}p^x(1 - p)^{n-x}, \quad x = 0, 1, 2, \ldots, n$

$M_X(t) = (pe^t + q)^n$

$M_X'(t) = n(pe^t + q)^{n-1}(pe^t)$

$\Rightarrow M_X'(0) = n(pe^0 + q)^{n-1}(pe^0)$

$= n(p + 1 - p)^{n-1}p = np$

PDF 09 / slide 46
$X \sim U(0, 1), \quad f(x) = 1, \quad x \in [0, 1]$

$m(t) = \int_0^1 e^{xt} \cdot 1\,dx$

$= \frac{1}{t}e^{xt}\Big|_{x=0}^{1} = \frac{e^t - 1}{t}, \quad t \ne 0$

$m(0) = 1$

PDF 09 / slide 48
$m_X(0) = 1$

$m_X(t) = E(e^{tX})$, hence $m_X(0) = E(e^{0 \cdot X}) = E(1) = 1$

$m_X(t) = 1 + \mu_1 t + \mu_2\frac{t^2}{2!} + \mu_3\frac{t^3}{3!} + \cdots + \mu_k\frac{t^k}{k!} + \cdots$

$m_X^{(k)}(0) = \left(\frac{d^k}{dt^k}m_X(t)\right)\Big|_{t=0} = \mu_k$

PDF 09 / slide 49
$X \sim U(0, 1), \quad f(x) = 1, \quad x \in [0, 1]$

$E(X) = \frac{0 + 1}{2} = \frac{1}{2} \qquad m(t) = \frac{e^t - 1}{t}, \quad t \ne 0$

$m(0) = 1$

$E(X) = m'(0) = \lim_{t \to 0}\frac{te^t - e^t + 1}{t^2} = \frac{1}{2}$

$t \to 0$

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